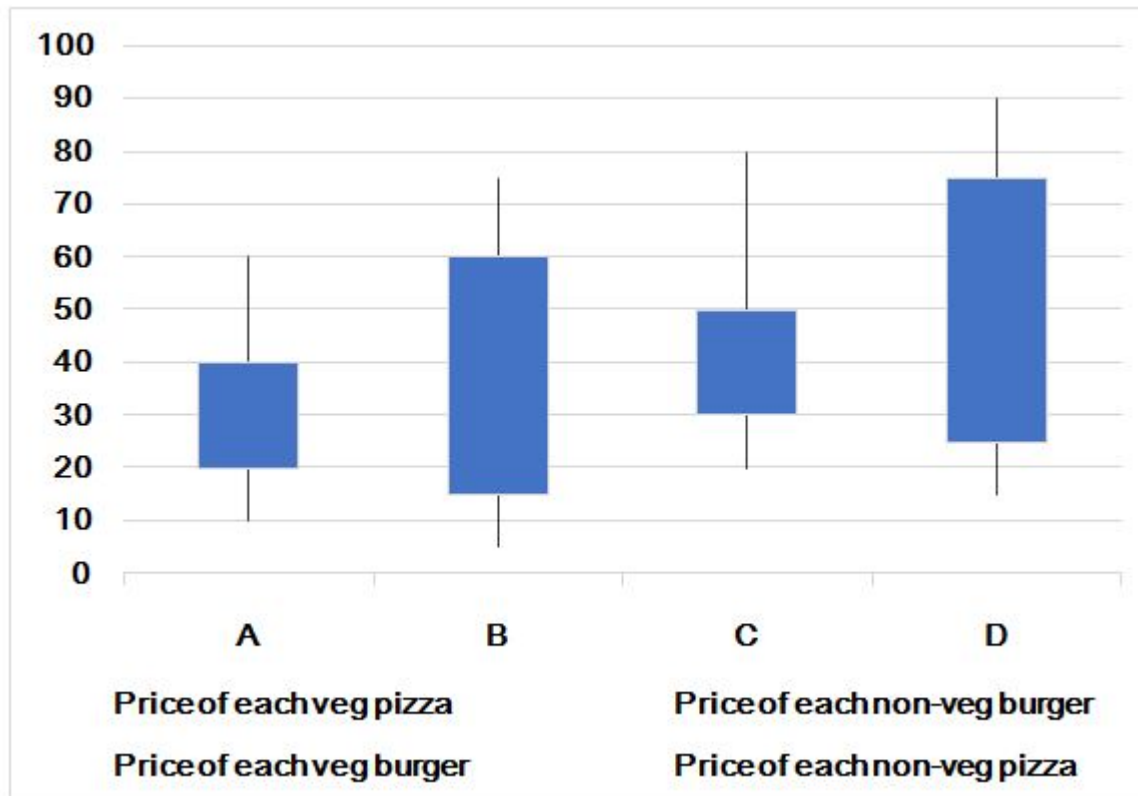


1. Questions

Read the following information carefully and answer the questions based on it.

The candle stick graph given below shows the price (in Rs.) of each veg and non-veg pizza and each veg and non-veg burger sold by four shops A, B, C, and D.



Interpretation of figures:

The floor and ceiling of the box represent the price of each veg pizza and each non-veg pizza respectively, while the floor and ceiling of the line represent the price of each veg burger and each non-veg burger sold by each shop.

Note:

The total number of items sold = the number of pizzas sold (veg + non-veg) + the number of burgers sold (veg + non-veg)

The total revenue generated = (the price of each item) * (the total number of items sold)

The total number of pizzas (veg + non-veg) sold by shops A, B, C, and D is 60, 50, 25, and 90 respectively.

The total number of burgers (veg + non-veg) sold by shops A, B, C and D is 35, 50, 60, and 40 respectively.

The total revenue generated from the sale of all pizzas (veg + non-veg) sold by shops A, B, C, and D is Rs. 1500, Rs. 2100, Rs. 950, and Rs. 4750 respectively.

The total revenue generated from the sale of all burgers (veg + non-veg) sold by shops A, B, C, and D is Rs. 850, Rs. 1300, Rs. 3600, and Rs. 2850 respectively.

Find the ratio of the number of veg pizzas sold by shop A and non-veg burgers sold by shop D.

- a. 3:2
- b. 4:1
- c. 5:3
- d. 2:1
- e. 1:1

2. Questions

The number of veg pizzas sold to males by shop B is 5 less than the number of non-veg pizzas sold to males by the same shop. If the ratio of the number of non-veg pizzas sold to males and females by shop B is 2:1, then find the number of veg pizzas sold to females by shop B.

- a. 10
- b. 15
- c. 12
- d. 5
- e. 8

3. Questions

Find the difference between the total revenue generated from the sale of all veg and non-veg burgers by shop C.

- a. Rs. 2000
- b. Rs. 3200
- c. Rs. 4000
- d. Rs. 1600
- e. Rs. 2800

4. Questions

In shop A, if the discount allowed on each veg burger is $\frac{X}{100}$ % of the actual price and the total number of veg burgers sold is also increased by $\frac{Y}{100}$ %, then the ratio of the price of each veg burger in shop A and the total number of burgers sold by the same shop will be 4:15. The total revenue generated from the sale of all veg burgers by shop A is reduced by Rs. 10.

- a. $X > Y$
- b. $X < Y$
- c. $X \geq Y$
- d. $X \leq Y$
- e. $X = Y$

5. Questions

Find the total number of veg burgers sold by all four shops together.

- 98
- 90
- 95
- 92
- 94

6. Questions

In shop E, the price of each veg pizza and each non-veg pizza is Rs. 10 and Rs. 20 respectively. If the number of veg pizzas sold by E is 20% less than that of shop D and the number of non-veg pizzas sold by E is $\frac{1}{5}$ th that of shop D, then find the total revenue generated from the sale of all pizzas (veg + non-veg) by shop E.

- Rs. 400
- Rs. 720
- Rs. 520
- Rs. 600
- Rs. 800

7. Questions

Three equations are given below.

Equation I: $A^2 - XA + 200 = 0$

Equation II: $B^2 - YB - 84 = 0$

Equation III: $C^2 + ZC - 21 = 0$

The smaller root of equation I is the second-lowest multiple of 5.

The positive root of equation II is exactly divisible by 7 but more than 10 and less than 20.

The value of $[Y * Z - X]$ is [?] number.

What should come in place of the question mark?

- Smallest composite
- Highest single-digit prime
- Even prime
- Odd prime

e. Odd composite

8. Questions

Two quadratic equations are given below.

Equation I: $PX^2 - 41X + 2R = 0$

Equation II: $QY^2 - 43Y + R = 0$

$Q > R > P$.

P, Q, and R are consecutive two-digit natural numbers, but neither of them is greater than 15.

Any two of either P, Q, or R are prime numbers.

Which of the following is/are correct?

I: The greater root of both equations is the same.

II: The smaller root of equation I is more than 1.

III: The difference between the roots of equation II is less than 3.

- a. Only II
- b. Both I and III
- c. Both II and III
- d. Only III
- e. All I, II and III

9. Questions

Two quadratic equations are given below.

Equation I: $P^2 - 19P + (23C + 1) = 0$

Equation II: $Q^2 - 28Q + 33D = 0$

The smaller root of equation I is $(C + 2)$.

The greater root of equation II is $(5D + 2)$.

Both C and D are positive integers.

Determine the correct relation between [C] and [D].

- a. $C > D$
- b. $C < D$
- c. $C \geq D$
- d. $C \leq D$

@QuizByMickey

e. $C = D$

10. Questions

An equation is given below.

Equation: $XT^2 + YT - Z = 0$

Y is a perfect square of a single-digit even composite number.

The sum of X and Y is equal to Z.

The difference between X and Z is more than 20 but less than 40.

The average of X and Z is 23.

Find the smaller root of the equation given above.

- a. -7.45
- b. -6.13
- c. -5.55
- d. -8.20
- e. -9.11

11. Questions

An equation is given below.

Equation: $K^2 - 35K + (MN + 45) = 0$

The smaller root is a single-digit prime number.

The greater root is an odd integer.

Note: $M > N$.

Find the value of 10M% of 20N.

- a. 36
- b. 20
- c. 18
- d. 55
- e. 42

@QuizByMickey

12. Questions

Read the following information carefully and answer the questions based on it.

The table chart given below shows the total number of voters registered (cast + non-cast) in an election in four different years 2004, 2009, 2014, and 2019. It also shows the percentage of the number of votes cast

out of the total number of voters registered and the number of valid votes cast each year.

Year	The total number of voters registered	The percentage of the number of votes cast	The number of valid votes cast
2004	$(16X + 80)$	60%	900
2009	3200	$(4Y - 5)\%$	$(Z + 300)$
2014	$80Y$	80%	W
2019	4000	$(X - 70)\%$	1400

Note:

The total number of voters registered = the number of votes cast+ the number of non-casted votes

The total number of votes cast= the number of valid votes cast + the number of invalid votes cast

The number of invalid votes cast in 2004 is half of 2019.

The number of non-cast votes in 2009 was 480 more than in 2014.

The average of the number of invalid votes cast in 2009 and 2014 is 440

The difference between the values of Z and W is $(7X + 3Y)$. ($Z > W$)

In 2009, if $\frac{1}{8}^{\text{th}}$ of the total number of votes cast (valid + invalid) are cast by females and the ratio of the number of valid votes cast by males and females is 9:1, then find the difference between the number of invalid votes cast by males and females in 2009.

- a. 300
- b. 120
- c. 180
- d. 150
- e. 200

13. Questions

Find the average number of votes cast in 2014 and 2019 together.

- a. 810
- b. 2560
- c. 490
- d. 1640
- e. 3200

14. Questions

The total number of votes registered in 2024 is $__\text{[?]}_\%$ less than in 2004 and the number of votes

cast in 2024 is 840. The ratio of the number of valid and invalid cast in 2024 is 9:5. In 2024, the number of non-cast votes is 60 more than the number of invalid votes cast.

What should come in place of the question mark?

- a. 40
- b. 32
- c. 55
- d. 24
- e. 18

15. Questions

Which of the following is/are correct?

I: $(W / Y) = 40$

II: $(X + 1580) = Z$

III: $(W - X) = (Z / Y)$

- a. Only II
- b. Both I and II
- c. Only III
- d. Both I and III
- e. All I, II and III

16. Questions

The following question has two series as series I and series II. Find the pattern of series I and series II in each question and answer accordingly.

Series I and II have one missing term as [P] and [Q] respectively. Both series follow a different pattern.

Series I: 24, 35, 13, 46, __[P]__, 57

Series II: 14, 29, 89, __[Q]__, 1799, 10799

@QuizByMickey

Which of the following is/are correct?

I: If [P] is added to [Q], then the resultant will be a perfect square.

II: Both [P] and [Q] are three-digit natural numbers.

III: [P] is an even prime number.

- a. Only I
- b. Both I and II

- c. Both I and III
- d. Only III
- e. All I, II and III

17. Questions

Series I has one wrong term and Series II starts with the correct term that will replace the wrong term of Series I. Both series follow the same pattern.

Series I: 7, 11, 36, 81, 221, 417

Series II: __[A]__, ____, ____, ____, ____, __[B]__

Find the greater root of the equation $L^2 - (\sqrt{A} + 1)L + (B / 17) = 0$.

- a. 5
- b. 6
- c. 3
- d. 7
- e. 4

18. Questions

Series I has two wrong terms. The correct terms that will replace the first and second wrong terms of series I are [X] and [Y] respectively. Series II does not follow the same pattern as Series I.

Series I: 24, 144, 16, 72, 6, 72

Series II: [X + Y], 24, 36, 90, 315, __[?]__

What should be the correct replacement for the question mark?

- a. 1714.50
- b. 1216.50
- c. 1417.50
- d. 1311.50
- e. 1819.50

@QuizByMickey

19. Questions

Series I and II have one wrong term each. Both series follow the same pattern.

Series I: [A], [3C + 2], 55, 94, 145, 211

Series II: [B - 20], 22, 48, 87, 139, 200

C is the highest single-digit perfect square.

The sum of A and B is 5 times of C.

The difference between A and B is 13. ($A < B$)

Find the difference between the correct terms that will replace the wrong terms of series I and II.

- a. 10
- b. 42
- c. 36
- d. 26
- e. 58

20. Questions

Series I has one missing term as [M] and series II has one wrong term. Both series follow the same pattern.

Series I: 100, 127, 2, __[M]__, - 384, 947

Series II: [M / 3], 140, 17, 360, - 369, 962

If [N] is the wrong term of series II, then determine the correct relation between [M] and [N].

- a. $M > N$
- b. $M < N$
- c. $M \geq N$
- d. $M \leq N$
- e. $M = N$

Explanations:

1. Questions

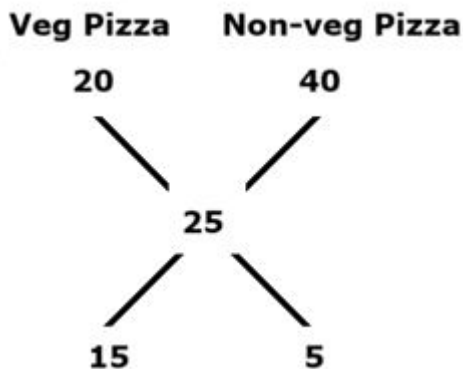
From the graph given above, we have,

Shop	The price of each veg pizza (in Rs.)	The price of each non-veg pizza (in Rs.)	The price of each veg burger (in Rs.)	The price of each non-veg burger (in Rs.)
A	20	40	10	60
B	15	60	5	75
C	30	50	20	80
D	25	75	15	90

Shop A

The overall price of each pizza (veg + non-veg) = $(1500 / 60) = \text{Rs. } 25$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg pizzas sold = $15:5 = 3:1$

The total number of veg pizzas sold = $3/4$ of $60 = 45$

So, the total revenue generated from all veg pizzas sold = $(20 * 45) = \text{Rs. } 900$

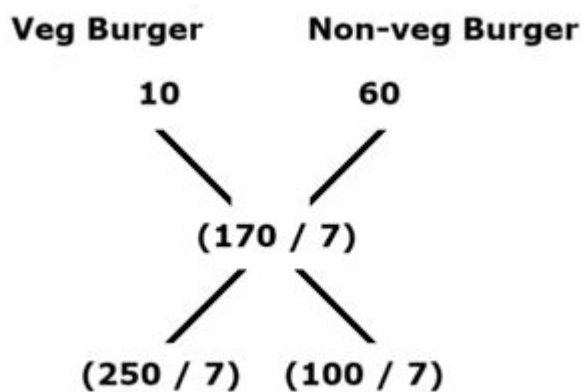
The total number of veg non-pizzas sold = $1/4$ of $60 = 15$

So, the total revenue generated from all non-veg pizzas sold = $(40 * 15) = \text{Rs. } 600$

Now,

The overall price of each burger (veg + non-veg) = $(850 / 35) = (170 / 7)$

So, using the allegation method, we get,



@QuizByMickey

So, the ratio of the number of veg and non-veg burgers sold = $(250 / 7) : (100 / 7) = 250:100 = 5:2$.

The total number of veg burgers sold = $5/7$ of $35 = 25$

So, the total revenue generated from the sale of all veg burgers = $(10 * 25) = \text{Rs. } 250$

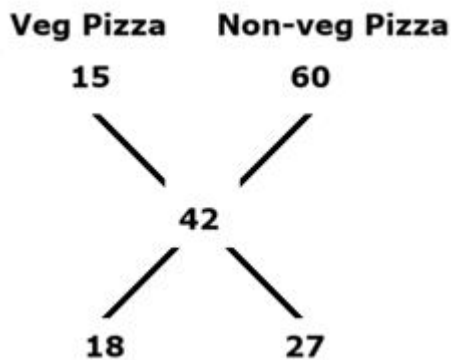
The total number of non-veg burgers sold = $2/7$ of $35 = 10$

So, the total revenue generated from the sale of all non-veg burgers = $(60 * 10) = \text{Rs. } 600$

Shop B

The overall price of each pizza (veg + non-veg) = $(2100 / 50) = \text{Rs. } 42$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg pizzas sold = $18:27 = 2:3$

The total number of veg pizzas sold = $2/5$ of $50 = 20$

So, the total revenue generated from all veg pizzas sold = $(15 * 20) = \text{Rs. } 300$

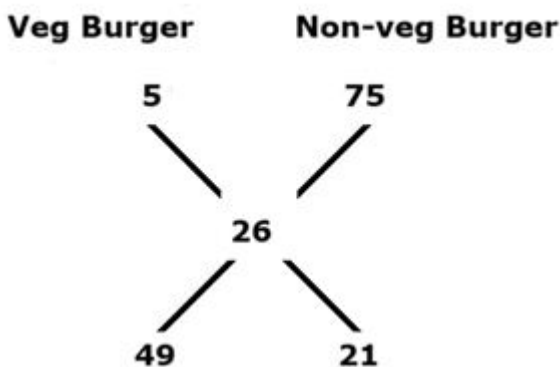
The total number of veg non-pizzas sold = $3/5$ of $50 = 30$

So, the total revenue generated from all non-veg pizzas sold = $(60 * 30) = \text{Rs. } 1800$

Now,

The overall price of each burger (veg + non-veg) = $(1300 / 50) = 26$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg burgers sold = $49:21 = 7:3$

The total number of veg burgers sold = $7/10$ of $50 = 35$

So, the total revenue generated from the sale of all veg burgers = $(5 * 35) = \text{Rs. } 175$

The total number of non-veg burgers sold = $3/10$ of $50 = 15$

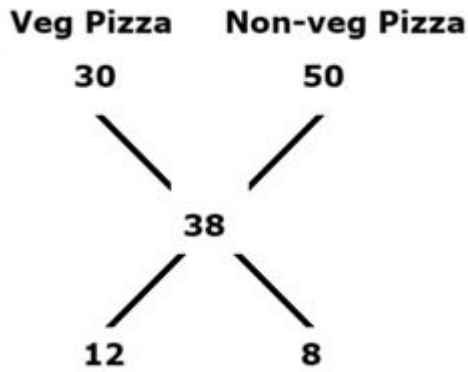
So, the total revenue generated from the sale of all non-veg burgers = $(15 * 75) = \text{Rs. } 1125$

Shop C

The overall price of each pizza (veg + non-veg) = $(950 / 25) = \text{Rs. } 38$

So, using the allegation method, we get,

@QuizByMickey



So, the ratio of the number of veg and non-veg pizzas sold = $12:8 = 3:2$

The total number of veg pizzas sold = $\frac{3}{5}$ of 25 = 15

So, the total revenue generated from all veg pizzas sold = $(30 * 15) = \text{Rs. } 450$

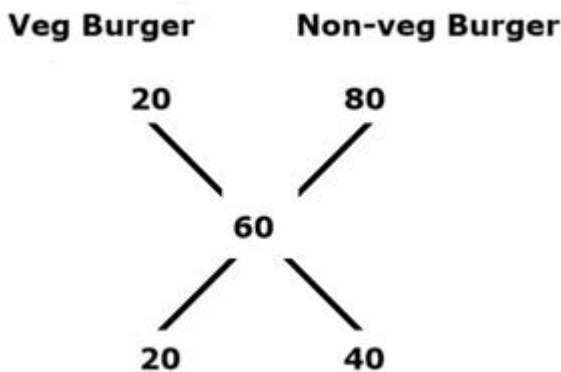
The total number of veg non-pizzas sold = $\frac{2}{5}$ of 25 = 10

So, the total revenue generated from all non-veg pizzas sold = $(50 * 10) = \text{Rs. } 500$

Now,

The overall price of each burger (veg + non-veg) = $(3600 / 60) = 60$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg burgers sold = $20:40 = 1:2$

The total number of veg burgers sold = $\frac{1}{3}$ of 60 = 20

So, the total revenue generated from the sale of all veg burgers = $(20 * 20) = \text{Rs. } 400$

The total number of non-veg burgers sold = $\frac{2}{3}$ of 60 = 40

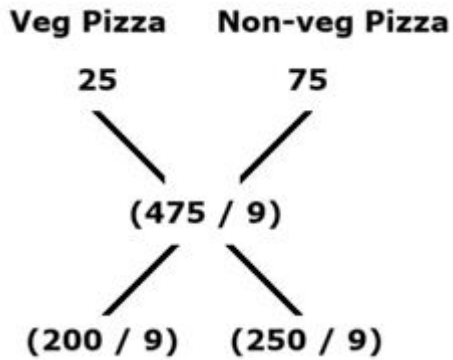
So, the total revenue generated from the sale of all non-veg burgers = $(80 * 40) = \text{Rs. } 3200$

Shop D

The overall price of each pizza (veg + non-veg) = $(4750 / 90) = \text{Rs. } (475 / 9)$

So, using the allegation method, we get,

@QuizByMickey



So, the ratio of the number of veg and non-veg pizzas sold = $(200 / 9) : (250 / 9) = 200:250 = 4:5$

The total number of veg pizzas sold = $4/9$ of $90 = 40$

So, the total revenue generated from all veg pizzas sold = $(25 * 40) = \text{Rs. } 1000$

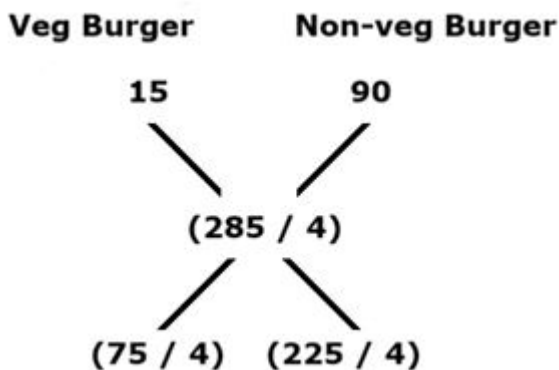
The total number of veg non-pizzas sold = $5/9$ of $90 = 50$

So, the total revenue generated from all non-veg pizzas sold = $(75 * 50) = \text{Rs. } 3750$

Now,

The overall price of each burger (veg + non-veg) = $(2850 / 40) = (285 / 4)$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg burgers sold = $(75 / 4) : (225 / 4) = 1:3$

The total number of veg burgers sold = $1/4$ of $40 = 10$

So, the total revenue generated from the sale of all veg burgers = $(15 * 10) = \text{Rs. } 150$

The total number of non-veg burgers sold = $3/4$ of $40 = 30$

@QuizByMickey

So, the total revenue generated from the sale of all non-veg burgers = $(90 * 30) = \text{Rs. } 2700$

The total number of items sold

Shop	Veg Pizza	Non-veg Pizza	Veg Burger	Non-veg Burger
A	45	15	25	10
B	20	30	35	15
C	15	10	20	40
D	40	50	10	30
Total	130	105	90	95

The total revenue generated (in Rs.)

Shop	Veg Pizza	Non-veg Pizza	Veg Burger	Non-veg Burger
A	900	600	250	600
B	300	1800	175	1125
C	450	500	400	3200
D	1000	3750	150	2700
Total	2650	6650	975	7625

Answer: A

Now, according to the question,

The number of veg pizzas sold by shop A = 45

The number of non-veg burgers sold by shop D = 30

Therefore, the required ratio = $45:30 = 2:1$.

Hence, the correct answer is option A.

2. Questions

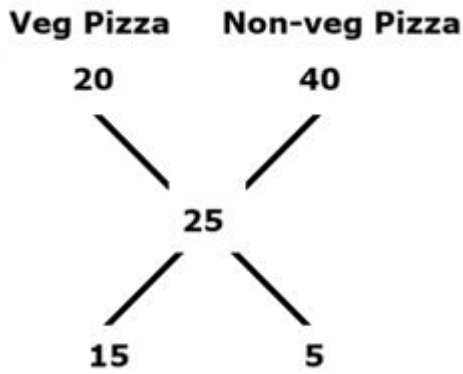
From the graph given above, we have,

Shop	The price of each veg pizza (in Rs.)	The price of each non-veg pizza (in Rs.)	The price of each veg burger (in Rs.)	The price of each non-veg burger (in Rs.)
A	20	40	10	60
B	15	60	5	75
C	30	50	20	80
D	25	75	15	90

Shop A

The overall price of each pizza (veg + non-veg) = $(1500 / 60) = \text{Rs. } 25$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg pizzas sold = 15:5 = 3:1

The total number of veg pizzas sold = $\frac{3}{4}$ of 60 = 45

So, the total revenue generated from all veg pizzas sold = $(20 * 45)$ = Rs. 900

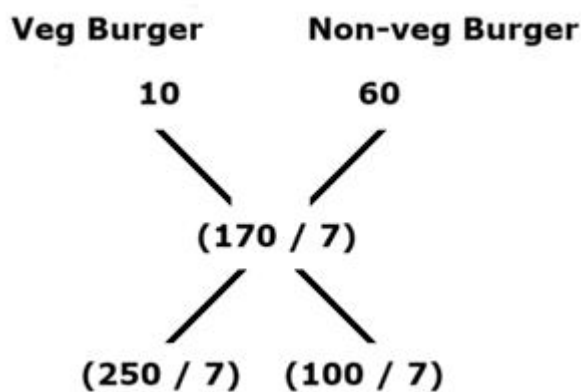
The total number of veg non-pizzas sold = $\frac{1}{4}$ of 60 = 15

So, the total revenue generated from all non-veg pizzas sold = $(40 * 15)$ = Rs. 600

Now,

The overall price of each burger (veg + non-veg) = $(850 / 35)$ = $(170 / 7)$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg burgers sold = $(250 / 7) : (100 / 7)$ = 250:100 = 5:2.

The total number of veg burgers sold = $\frac{5}{7}$ of 35 = 25

So, the total revenue generated from the sale of all veg burgers = $(10 * 25)$ = Rs. 250

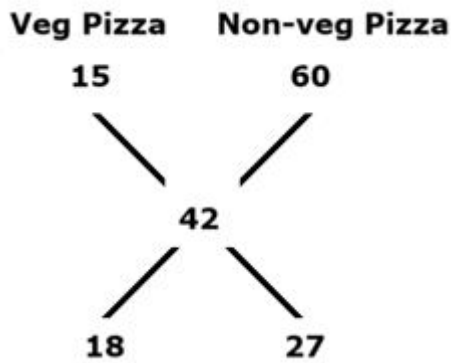
The total number of non-veg burgers sold = $\frac{2}{7}$ of 35 = 10

So, the total revenue generated from the sale of all non-veg burgers = $(60 * 10)$ = Rs. 600

Shop B

The overall price of each pizza (veg + non-veg) = $(2100 / 50)$ = Rs. 42

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg pizzas sold = $18:27 = 2:3$

The total number of veg pizzas sold = $\frac{2}{5}$ of 50 = 20

So, the total revenue generated from all veg pizzas sold = $(15 * 20) = \text{Rs. } 300$

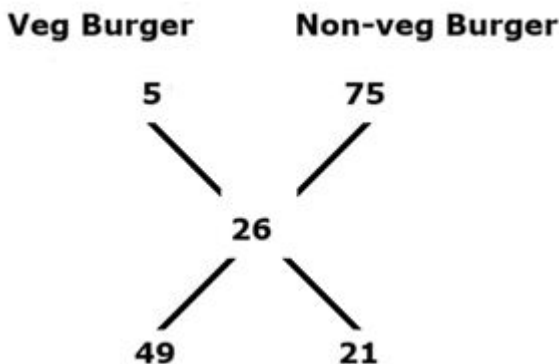
The total number of veg non-pizzas sold = $\frac{3}{5}$ of 50 = 30

So, the total revenue generated from all non-veg pizzas sold = $(60 * 30) = \text{Rs. } 1800$

Now,

The overall price of each burger (veg + non-veg) = $(1300 / 50) = 26$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg burgers sold = $49:21 = 7:3$

The total number of veg burgers sold = $\frac{7}{10}$ of 50 = 35

So, the total revenue generated from the sale of all veg burgers = $(5 * 35) = \text{Rs. } 175$

The total number of non-veg burgers sold = $\frac{3}{10}$ of 50 = 15

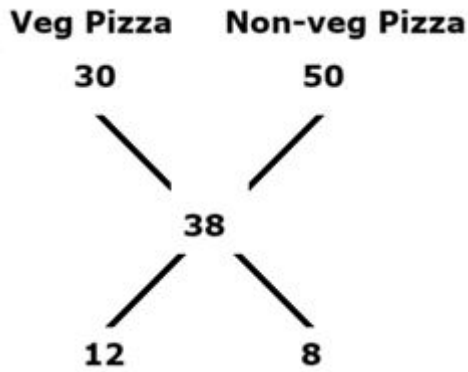
@QuizByMickey

So, the total revenue generated from the sale of all non-veg burgers = $(15 * 75) = \text{Rs. } 1125$

Shop C

The overall price of each pizza (veg + non-veg) = $(950 / 25) = \text{Rs. } 38$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg pizzas sold = $12:8 = 3:2$

The total number of veg pizzas sold = $\frac{3}{5}$ of 25 = 15

So, the total revenue generated from all veg pizzas sold = $(30 * 15) = \text{Rs. } 450$

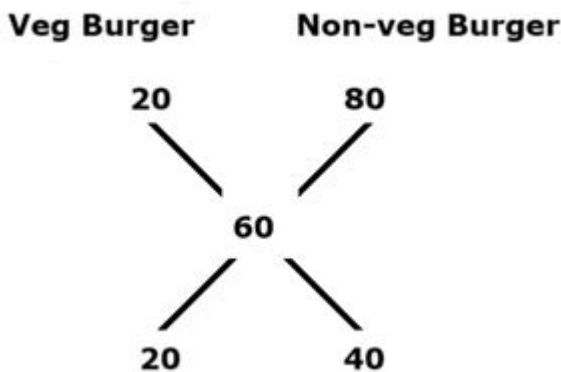
The total number of veg non-pizzas sold = $\frac{2}{5}$ of 25 = 10

So, the total revenue generated from all non-veg pizzas sold = $(50 * 10) = \text{Rs. } 500$

Now,

The overall price of each burger (veg + non-veg) = $(3600 / 60) = 60$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg burgers sold = $20:40 = 1:2$

The total number of veg burgers sold = $\frac{1}{3}$ of 60 = 20

So, the total revenue generated from the sale of all veg burgers = $(20 * 20) = \text{Rs. } 400$

The total number of non-veg burgers sold = $\frac{2}{3}$ of 60 = 40

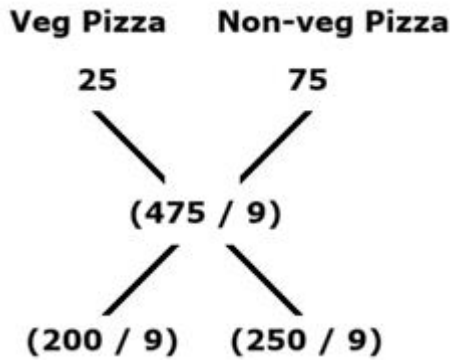
So, the total revenue generated from the sale of all non-veg burgers = $(80 * 40) = \text{Rs. } 3200$

Shop D

The overall price of each pizza (veg + non-veg) = $(4750 / 90) = \text{Rs. } (475 / 9)$

So, using the allegation method, we get,

@QuizByMickey



So, the ratio of the number of veg and non-veg pizzas sold = $(200 / 9) : (250 / 9) = 200:250 = 4:5$

The total number of veg pizzas sold = $4/9$ of $90 = 40$

So, the total revenue generated from all veg pizzas sold = $(25 * 40) = \text{Rs. } 1000$

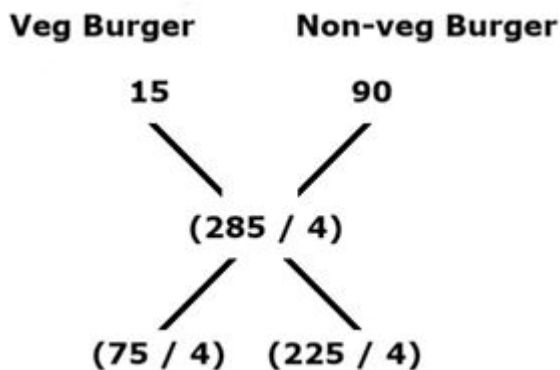
The total number of veg non-pizzas sold = $5/9$ of $90 = 50$

So, the total revenue generated from all non-veg pizzas sold = $(75 * 50) = \text{Rs. } 3750$

Now,

The overall price of each burger (veg + non-veg) = $(2850 / 40) = (285 / 4)$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg burgers sold = $(75 / 4) : (225 / 4) = 1:3$

The total number of veg burgers sold = $1/4$ of $40 = 10$

So, the total revenue generated from the sale of all veg burgers = $(15 * 10) = \text{Rs. } 150$

The total number of non-veg burgers sold = $3/4$ of $40 = 30$

@QuizByMickey

So, the total revenue generated from the sale of all non-veg burgers = $(90 * 30) = \text{Rs. } 2700$

The total number of items sold

Shop	Veg Pizza	Non-veg Pizza	Veg Burger	Non-veg Burger
A	45	15	25	10
B	20	30	35	15
C	15	10	20	40
D	40	50	10	30
Total	130	105	90	95

The total revenue generated (in Rs.)

Shop	Veg Pizza	Non-veg Pizza	Veg Burger	Non-veg Burger
A	900	600	250	600
B	300	1800	175	1125
C	450	500	400	3200
D	1000	3750	150	2700
Total	2650	6650	975	7625

Answer: D

Shop B

The total number of non-veg pizzas sold = 30

The number of non-veg pizzas sold to males = $\frac{2}{3}$ of 30 = 20

The number of non-veg pizzas sold to females = $\frac{1}{3}$ of 30 = 10

The total number of veg pizzas sold = 20

The number of veg pizzas sold to males = $(20 - 5) = 15$

Therefore, the number of veg pizzas sold to females by shop B = $(20 - 15) = 5$.

Hence, the correct answer is option D.

3. Questions

From the graph given above, we have,

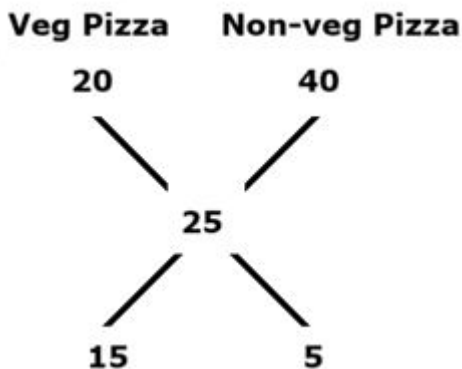
Shop	The price of each veg pizza (in Rs.)	The price of each non-veg pizza (in Rs.)	The price of each veg burger (in Rs.)	The price of each non-veg burger (in Rs.)
A	20	40	10	60
B	15	60	5	75
C	30	50	20	80
D	25	75	15	90

@QuizByMickey

Shop A

The overall price of each pizza (veg + non-veg) = $(1500 / 60) = \text{Rs. } 25$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg pizzas sold = $15:5 = 3:1$

The total number of veg pizzas sold = $3/4$ of $60 = 45$

So, the total revenue generated from all veg pizzas sold = $(20 * 45) = \text{Rs. } 900$

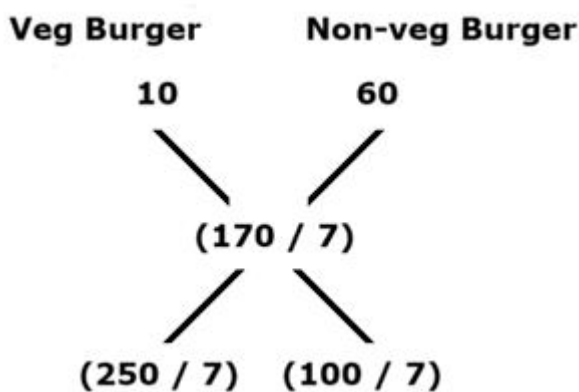
The total number of veg non-pizzas sold = $1/4$ of $60 = 15$

So, the total revenue generated from all non-veg pizzas sold = $(40 * 15) = \text{Rs. } 600$

Now,

The overall price of each burger (veg + non-veg) = $(850 / 35) = (170 / 7)$

So, using the allegation method, we get,



@QuizByMickey

So, the ratio of the number of veg and non-veg burgers sold = $(250 / 7) : (100 / 7) = 250:100 = 5:2$.

The total number of veg burgers sold = $5/7$ of $35 = 25$

So, the total revenue generated from the sale of all veg burgers = $(10 * 25) = \text{Rs. } 250$

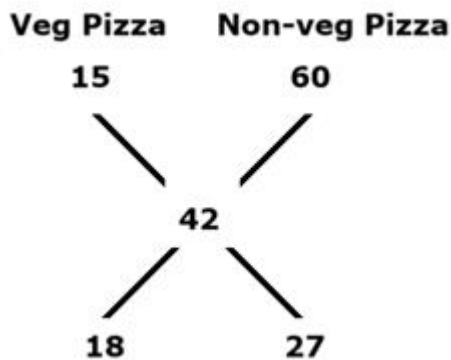
The total number of non-veg burgers sold = $2/7$ of $35 = 10$

So, the total revenue generated from the sale of all non-veg burgers = $(60 * 10) = \text{Rs. } 600$

Shop B

The overall price of each pizza (veg + non-veg) = $(2100 / 50) = \text{Rs. } 42$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg pizzas sold = $18:27 = 2:3$

The total number of veg pizzas sold = $2/5$ of $50 = 20$

So, the total revenue generated from all veg pizzas sold = $(15 * 20) = \text{Rs. } 300$

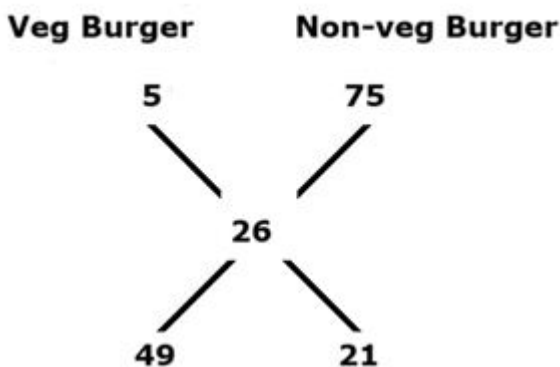
The total number of veg non-pizzas sold = $3/5$ of $50 = 30$

So, the total revenue generated from all non-veg pizzas sold = $(60 * 30) = \text{Rs. } 1800$

Now,

The overall price of each burger (veg + non-veg) = $(1300 / 50) = 26$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg burgers sold = $49:21 = 7:3$

The total number of veg burgers sold = $7/10$ of $50 = 35$

So, the total revenue generated from the sale of all veg burgers = $(5 * 35) = \text{Rs. } 175$

The total number of non-veg burgers sold = $3/10$ of $50 = 15$

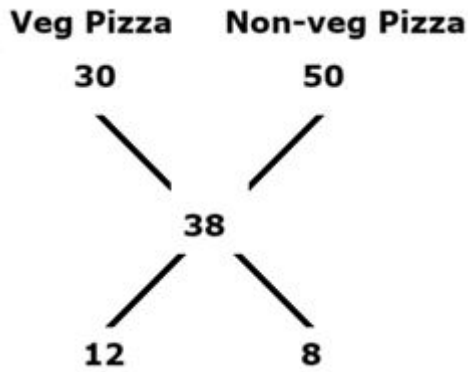
So, the total revenue generated from the sale of all non-veg burgers = $(15 * 75) = \text{Rs. } 1125$

Shop C

The overall price of each pizza (veg + non-veg) = $(950 / 25) = \text{Rs. } 38$

So, using the allegation method, we get,

@QuizByMickey



So, the ratio of the number of veg and non-veg pizzas sold = $12:8 = 3:2$

The total number of veg pizzas sold = $\frac{3}{5}$ of 25 = 15

So, the total revenue generated from all veg pizzas sold = $(30 * 15) = \text{Rs. } 450$

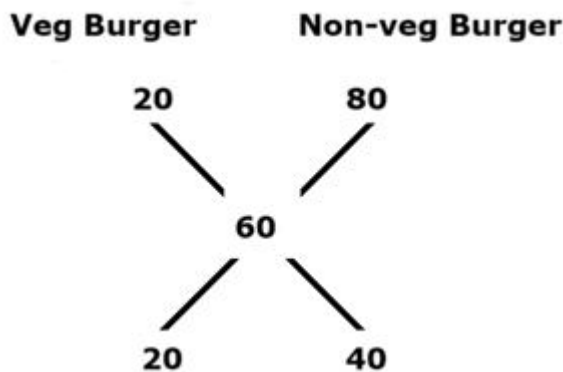
The total number of veg non-pizzas sold = $\frac{2}{5}$ of 25 = 10

So, the total revenue generated from all non-veg pizzas sold = $(50 * 10) = \text{Rs. } 500$

Now,

The overall price of each burger (veg + non-veg) = $(3600 / 60) = 60$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg burgers sold = $20:40 = 1:2$

The total number of veg burgers sold = $\frac{1}{3}$ of 60 = 20

So, the total revenue generated from the sale of all veg burgers = $(20 * 20) = \text{Rs. } 400$

The total number of non-veg burgers sold = $\frac{2}{3}$ of 60 = 40

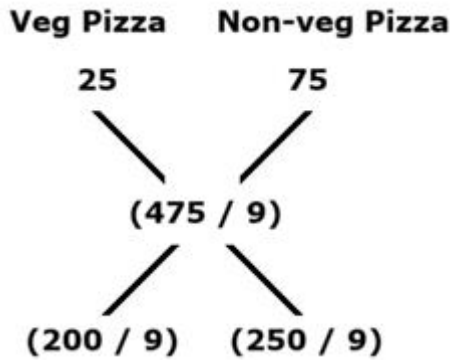
So, the total revenue generated from the sale of all non-veg burgers = $(80 * 40) = \text{Rs. } 3200$

Shop D

The overall price of each pizza (veg + non-veg) = $(4750 / 90) = \text{Rs. } (475 / 9)$

So, using the allegation method, we get,

@QuizByMickey



So, the ratio of the number of veg and non-veg pizzas sold = $(200 / 9) : (250 / 9) = 200:250 = 4:5$

The total number of veg pizzas sold = $4/9$ of $90 = 40$

So, the total revenue generated from all veg pizzas sold = $(25 * 40) = \text{Rs. } 1000$

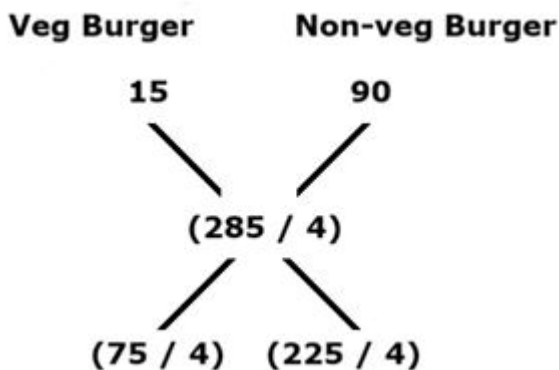
The total number of veg non-pizzas sold = $5/9$ of $90 = 50$

So, the total revenue generated from all non-veg pizzas sold = $(75 * 50) = \text{Rs. } 3750$

Now,

The overall price of each burger (veg + non-veg) = $(2850 / 40) = (285 / 4)$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg burgers sold = $(75 / 4) : (225 / 4) = 1:3$

The total number of veg burgers sold = $1/4$ of $40 = 10$

So, the total revenue generated from the sale of all veg burgers = $(15 * 10) = \text{Rs. } 150$

The total number of non-veg burgers sold = $3/4$ of $40 = 30$

@QuizByMickey

So, the total revenue generated from the sale of all non-veg burgers = $(90 * 30) = \text{Rs. } 2700$

The total number of items sold

Shop	Veg Pizza	Non-veg Pizza	Veg Burger	Non-veg Burger
A	45	15	25	10
B	20	30	35	15
C	15	10	20	40
D	40	50	10	30
Total	130	105	90	95

The total revenue generated (in Rs.)

Shop	Veg Pizza	Non-veg Pizza	Veg Burger	Non-veg Burger
A	900	600	250	600
B	300	1800	175	1125
C	450	500	400	3200
D	1000	3750	150	2700
Total	2650	6650	975	7625

Answer: E

Shop C

The total revenue generated from the sale of all veg burgers = Rs. 400

The total revenue generated from the sale of all non-veg burgers = Rs. 3200

Therefore, the required difference = $(3200 - 400) = \text{Rs. } 2800$.

Hence, the correct answer is option E.

4. Questions

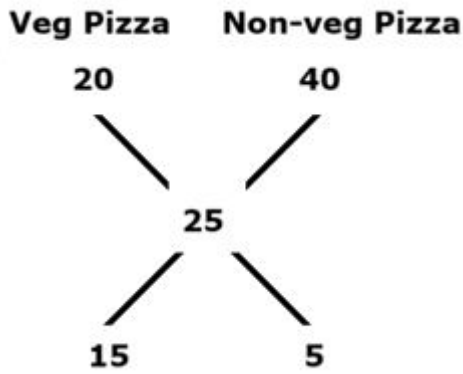
From the graph given above, we have,

Shop	The price of each veg pizza (in Rs.)	The price of each non-veg pizza (in Rs.)	The price of each veg burger (in Rs.)	The price of each non-veg burger (in Rs.)
A	20	40	10	60
B	15	60	5	75
C	30	50	20	80
D	25	75	15	90

Shop A

The overall price of each pizza (veg + non-veg) = $(1500 / 60) = \text{Rs. } 25$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg pizzas sold = 15:5 = 3:1

The total number of veg pizzas sold = $\frac{3}{4}$ of 60 = 45

So, the total revenue generated from all veg pizzas sold = $(20 * 45) = \text{Rs. } 900$

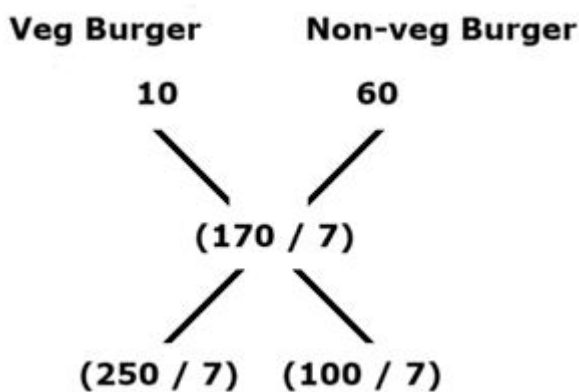
The total number of veg non-pizzas sold = $\frac{1}{4}$ of 60 = 15

So, the total revenue generated from all non-veg pizzas sold = $(40 * 15) = \text{Rs. } 600$

Now,

The overall price of each burger (veg + non-veg) = $(850 / 35) = (170 / 7)$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg burgers sold = $(250 / 7) : (100 / 7) = 250:100 = 5:2$.

The total number of veg burgers sold = $\frac{5}{7}$ of 35 = 25

So, the total revenue generated from the sale of all veg burgers = $(10 * 25) = \text{Rs. } 250$

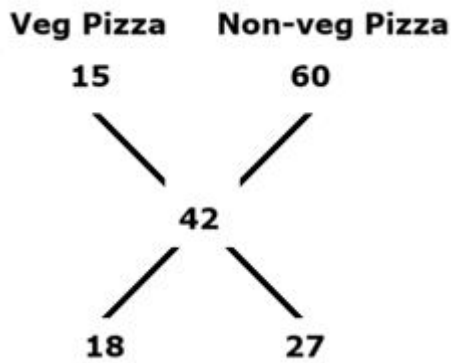
The total number of non-veg burgers sold = $\frac{2}{7}$ of 35 = 10

So, the total revenue generated from the sale of all non-veg burgers = $(60 * 10) = \text{Rs. } 600$

Shop B

The overall price of each pizza (veg + non-veg) = $(2100 / 50) = \text{Rs. } 42$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg pizzas sold = $18:27 = 2:3$

The total number of veg pizzas sold = $\frac{2}{5}$ of 50 = 20

So, the total revenue generated from all veg pizzas sold = $(15 * 20) = \text{Rs. } 300$

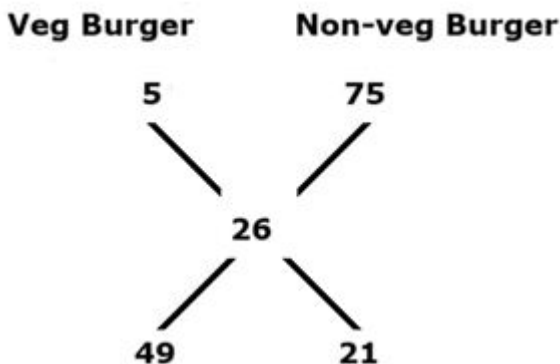
The total number of veg non-pizzas sold = $\frac{3}{5}$ of 50 = 30

So, the total revenue generated from all non-veg pizzas sold = $(60 * 30) = \text{Rs. } 1800$

Now,

The overall price of each burger (veg + non-veg) = $(1300 / 50) = 26$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg burgers sold = $49:21 = 7:3$

The total number of veg burgers sold = $\frac{7}{10}$ of 50 = 35

So, the total revenue generated from the sale of all veg burgers = $(5 * 35) = \text{Rs. } 175$

The total number of non-veg burgers sold = $\frac{3}{10}$ of 50 = 15

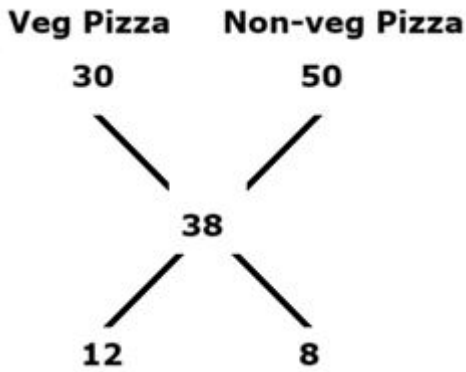
@QuizByMickey

So, the total revenue generated from the sale of all non-veg burgers = $(15 * 75) = \text{Rs. } 1125$

Shop C

The overall price of each pizza (veg + non-veg) = $(950 / 25) = \text{Rs. } 38$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg pizzas sold = $12:8 = 3:2$

The total number of veg pizzas sold = $\frac{3}{5}$ of 25 = 15

So, the total revenue generated from all veg pizzas sold = $(30 * 15) = \text{Rs. } 450$

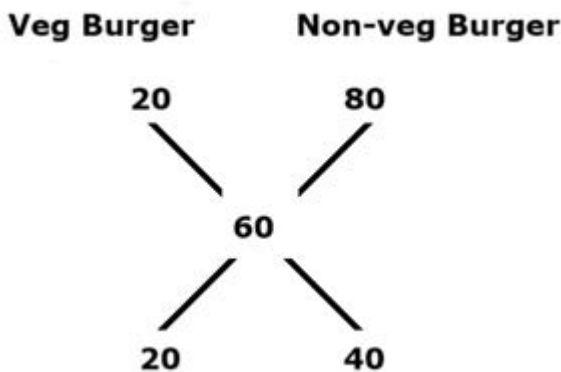
The total number of veg non-pizzas sold = $\frac{2}{5}$ of 25 = 10

So, the total revenue generated from all non-veg pizzas sold = $(50 * 10) = \text{Rs. } 500$

Now,

The overall price of each burger (veg + non-veg) = $(3600 / 60) = 60$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg burgers sold = $20:40 = 1:2$

The total number of veg burgers sold = $\frac{1}{3}$ of 60 = 20

So, the total revenue generated from the sale of all veg burgers = $(20 * 20) = \text{Rs. } 400$

The total number of non-veg burgers sold = $\frac{2}{3}$ of 60 = 40

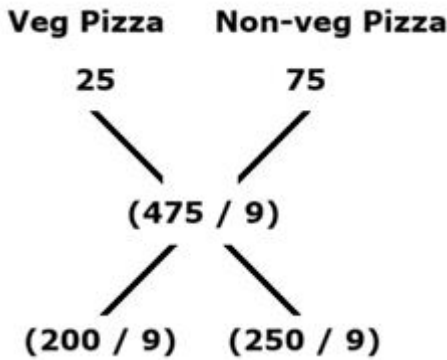
So, the total revenue generated from the sale of all non-veg burgers = $(80 * 40) = \text{Rs. } 3200$

Shop D

The overall price of each pizza (veg + non-veg) = $(4750 / 90) = \text{Rs. } (475 / 9)$

So, using the allegation method, we get,

@QuizByMickey



So, the ratio of the number of veg and non-veg pizzas sold = $(200 / 9) : (250 / 9) = 200:250 = 4:5$

The total number of veg pizzas sold = $4/9$ of $90 = 40$

So, the total revenue generated from all veg pizzas sold = $(25 * 40) = \text{Rs. } 1000$

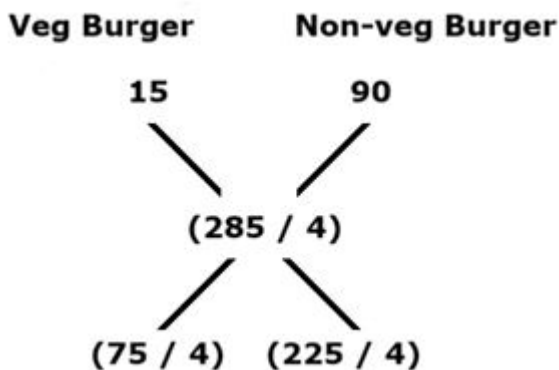
The total number of veg non-pizzas sold = $5/9$ of $90 = 50$

So, the total revenue generated from all non-veg pizzas sold = $(75 * 50) = \text{Rs. } 3750$

Now,

The overall price of each burger (veg + non-veg) = $(2850 / 40) = (285 / 4)$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg burgers sold = $(75 / 4) : (225 / 4) = 1:3$

The total number of veg burgers sold = $1/4$ of $40 = 10$

So, the total revenue generated from the sale of all veg burgers = $(15 * 10) = \text{Rs. } 150$

The total number of non-veg burgers sold = $3/4$ of $40 = 30$

@QuizByMickey

So, the total revenue generated from the sale of all non-veg burgers = $(90 * 30) = \text{Rs. } 2700$

The total number of items sold

Shop	Veg Pizza	Non-veg Pizza	Veg Burger	Non-veg Burger
A	45	15	25	10
B	20	30	35	15
C	15	10	20	40
D	40	50	10	30
Total	130	105	90	95

The total revenue generated (in Rs.)

Shop	Veg Pizza	Non-veg Pizza	Veg Burger	Non-veg Burger
A	900	600	250	600
B	300	1800	175	1125
C	450	500	400	3200
D	1000	3750	150	2700
Total	2650	6650	975	7625

Answer: E

Shop A

The new ratio of the price of each veg burger and the total number of veg burgers sold = 4:15. Let it be Rs. 4a and 15a respectively.

The total revenue generated = (250 – 10) = Rs. 240

Now,

$$4a * 15a = 240$$

$$60a^2 = 240$$

$$a^2 = 4$$

$$a = \sqrt{4}$$

$$a = 2$$

So, the new price of each veg burger = 4a = (4 * 2) = Rs. 8

And, the price of each veg burger initially = Rs. 10

The discount allowed on each veg burger = (10 – 8) = Rs. 2

$$X = 2/10 * 100 = 20.$$

Also, the increased number of veg burgers sold = 15a = (15 * 2) = 30

And, the number of veg burgers sold initially = 25

$$Y = (30 - 25) / 25 * 100 = 5/25 * 100 = 20.$$

@QuizByMickey

Therefore, $X = Y$.

Hence, the correct answer is option E.

5. Questions

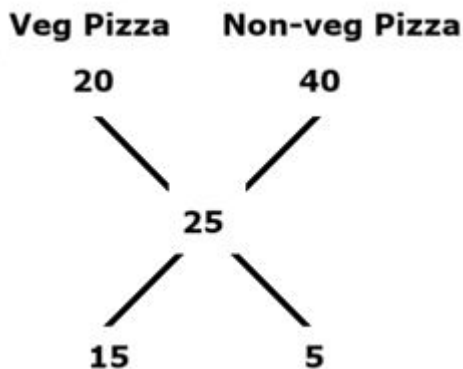
From the graph given above, we have,

Shop	The price of each veg pizza (in Rs.)	The price of each non-veg pizza (in Rs.)	The price of each veg burger (in Rs.)	The price of each non-veg burger (in Rs.)
A	20	40	10	60
B	15	60	5	75
C	30	50	20	80
D	25	75	15	90

Shop A

The overall price of each pizza (veg + non-veg) = $(1500 / 60) = \text{Rs. } 25$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg pizzas sold = $15:5 = 3:1$

The total number of veg pizzas sold = $3/4$ of $60 = 45$

So, the total revenue generated from all veg pizzas sold = $(20 * 45) = \text{Rs. } 900$

The total number of veg non-pizzas sold = $1/4$ of $60 = 15$

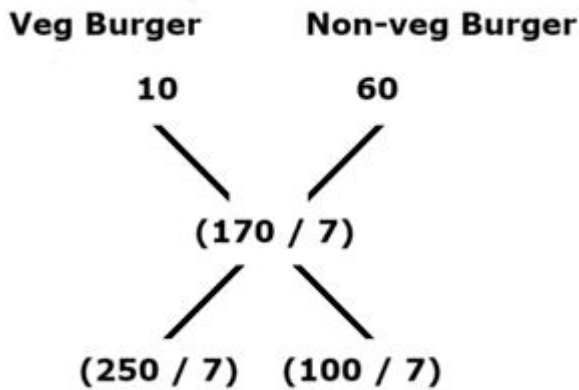
@QuizByMickey

So, the total revenue generated from all non-veg pizzas sold = $(40 * 15) = \text{Rs. } 600$

Now,

The overall price of each burger (veg + non-veg) = $(850 / 35) = (170 / 7)$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg burgers sold = $(250 / 7) : (100 / 7) = 250:100 = 5:2$.

The total number of veg burgers sold = $5/7$ of 35 = 25

So, the total revenue generated from the sale of all veg burgers = $(10 * 25) = \text{Rs. } 250$

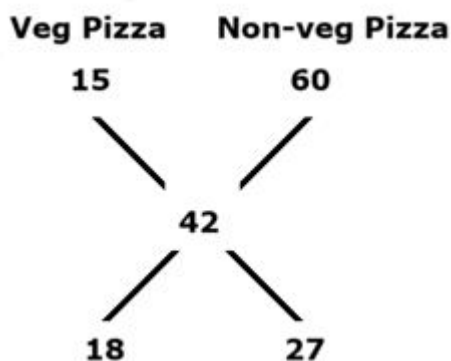
The total number of non-veg burgers sold = $2/7$ of 35 = 10

So, the total revenue generated from the sale of all non-veg burgers = $(60 * 10) = \text{Rs. } 600$

Shop B

The overall price of each pizza (veg + non-veg) = $(2100 / 50) = \text{Rs. } 42$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg pizzas sold = $18:27 = 2:3$

The total number of veg pizzas sold = $2/5$ of 50 = 20

So, the total revenue generated from all veg pizzas sold = $(15 * 20) = \text{Rs. } 300$

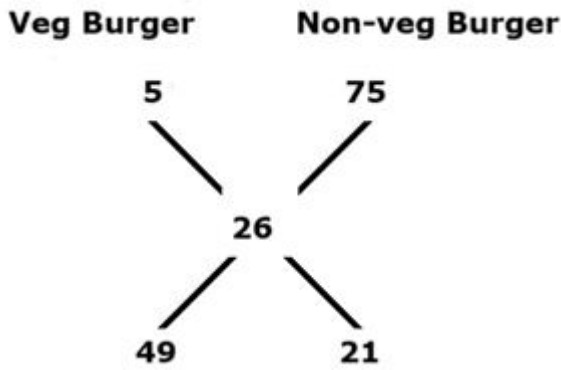
The total number of veg non-pizzas sold = $3/5$ of 50 = 30

So, the total revenue generated from all non-veg pizzas sold = $(60 * 30) = \text{Rs. } 1800$

Now,

The overall price of each burger (veg + non-veg) = $(1300 / 50) = 26$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg burgers sold = $49:21 = 7:3$

The total number of veg burgers sold = $7/10$ of $50 = 35$

So, the total revenue generated from the sale of all veg burgers = $(5 * 35) = \text{Rs. } 175$

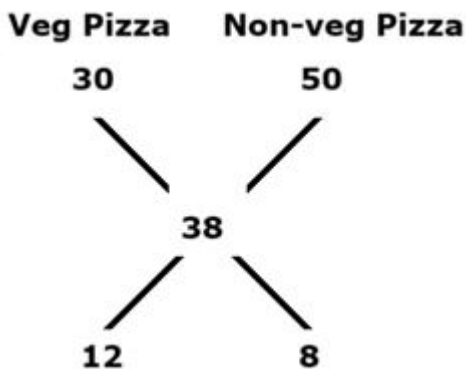
The total number of non-veg burgers sold = $3/10$ of $50 = 15$

So, the total revenue generated from the sale of all non-veg burgers = $(15 * 75) = \text{Rs. } 1125$

Shop C

The overall price of each pizza (veg + non-veg) = $(950 / 25) = \text{Rs. } 38$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg pizzas sold = $12:8 = 3:2$

The total number of veg pizzas sold = $3/5$ of $25 = 15$

So, the total revenue generated from all veg pizzas sold = $(30 * 15) = \text{Rs. } 450$

The total number of veg non-pizzas sold = $2/5$ of $25 = 10$

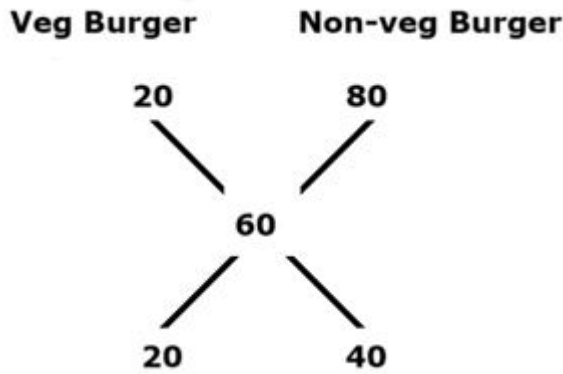
@QuizByMickey

So, the total revenue generated from all non-veg pizzas sold = $(50 * 10) = \text{Rs. } 500$

Now,

The overall price of each burger (veg + non-veg) = $(3600 / 60) = 60$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg burgers sold = $20:40 = 1:2$

The total number of veg burgers sold = $1/3$ of $60 = 20$

So, the total revenue generated from the sale of all veg burgers = $(20 * 20) = \text{Rs. } 400$

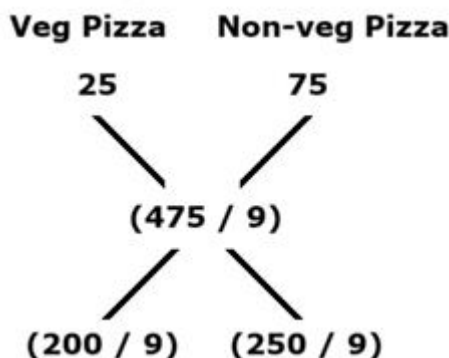
The total number of non-veg burgers sold = $2/3$ of $60 = 40$

So, the total revenue generated from the sale of all non-veg burgers = $(80 * 40) = \text{Rs. } 3200$

Shop D

The overall price of each pizza (veg + non-veg) = $(4750 / 90) = \text{Rs. } (475 / 9)$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg pizzas sold = $(200 / 9) : (250 / 9) = 200:250 = 4:5$

The total number of veg pizzas sold = $4/9$ of $90 = 40$

So, the total revenue generated from all veg pizzas sold = $(25 * 40) = \text{Rs. } 1000$

The total number of veg non-pizzas sold = $5/9$ of $90 = 50$

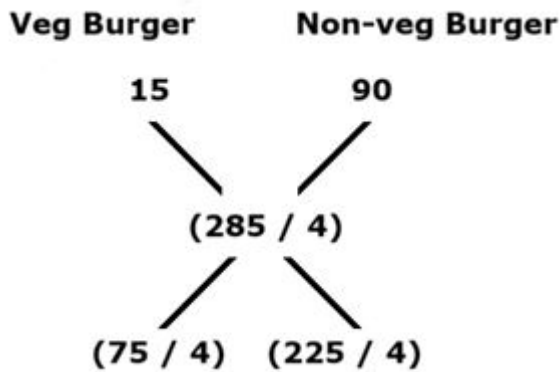
@QuizByMickey

So, the total revenue generated from all non-veg pizzas sold = $(75 * 50) = \text{Rs. } 3750$

Now,

The overall price of each burger (veg + non-veg) = $(2850 / 40) = (285 / 4)$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg burgers sold = $(75 / 4) : (225 / 4) = 1:3$

The total number of veg burgers sold = $1/4$ of 40 = 10

So, the total revenue generated from the sale of all veg burgers = $(15 * 10) = \text{Rs. } 150$

The total number of non-veg burgers sold = $3/4$ of 40 = 30

So, the total revenue generated from the sale of all non-veg burgers = $(90 * 30) = \text{Rs. } 2700$

The total number of items sold

Shop	Veg Pizza	Non-veg Pizza	Veg Burger	Non-veg Burger
A	45	15	25	10
B	20	30	35	15
C	15	10	20	40
D	40	50	10	30
Total	130	105	90	95

The total revenue generated (in Rs.)

Shop	Veg Pizza	Non-veg Pizza	Veg Burger	Non-veg Burger
A	900	600	250	600
B	300	1800	175	1125
C	450	500	400	3200
D	1000	3750	150	2700
Total	2650	6650	975	7625

Answer: B

The number of veg burgers sold by shop,

A = 25

B = 35

C = 20

D = 10

Therefore, the total number of veg burgers sold by all four shops together = $(25 + 35 + 20 + 10) = 90$.

Hence, the correct answer is option B.

6. Questions

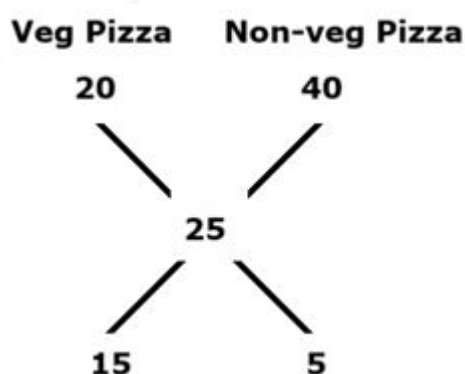
From the graph given above, we have,

Shop	The price of each veg pizza (in Rs.)	The price of each non-veg pizza (in Rs.)	The price of each veg burger (in Rs.)	The price of each non-veg burger (in Rs.)
A	20	40	10	60
B	15	60	5	75
C	30	50	20	80
D	25	75	15	90

Shop A

The overall price of each pizza (veg + non-veg) = $(1500 / 60) = \text{Rs. } 25$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg pizzas sold = $15:5 = 3:1$

The total number of veg pizzas sold = $3/4$ of $60 = 45$

So, the total revenue generated from all veg pizzas sold = $(20 * 45) = \text{Rs. } 900$

The total number of veg non-pizzas sold = $1/4$ of $60 = 15$

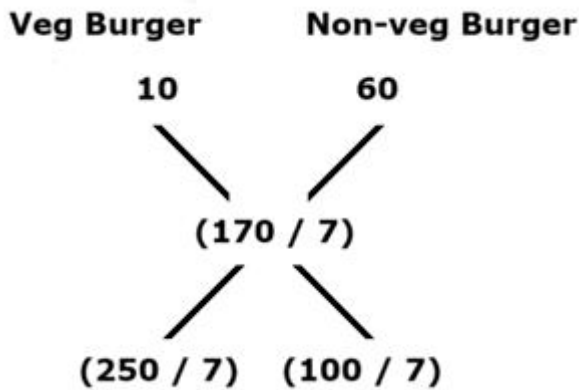
@QuizByMickey

So, the total revenue generated from all non-veg pizzas sold = $(40 * 15) = \text{Rs. } 600$

Now,

The overall price of each burger (veg + non-veg) = $(850 / 35) = (170 / 7)$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg burgers sold = $(250 / 7) : (100 / 7) = 250:100 = 5:2$.

The total number of veg burgers sold = $5/7$ of 35 = 25

So, the total revenue generated from the sale of all veg burgers = $(10 * 25) = \text{Rs. } 250$

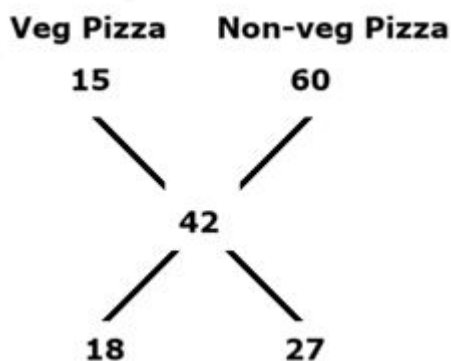
The total number of non-veg burgers sold = $2/7$ of 35 = 10

So, the total revenue generated from the sale of all non-veg burgers = $(60 * 10) = \text{Rs. } 600$

Shop B

The overall price of each pizza (veg + non-veg) = $(2100 / 50) = \text{Rs. } 42$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg pizzas sold = $18:27 = 2:3$

The total number of veg pizzas sold = $2/5$ of 50 = 20

So, the total revenue generated from all veg pizzas sold = $(15 * 20) = \text{Rs. } 300$

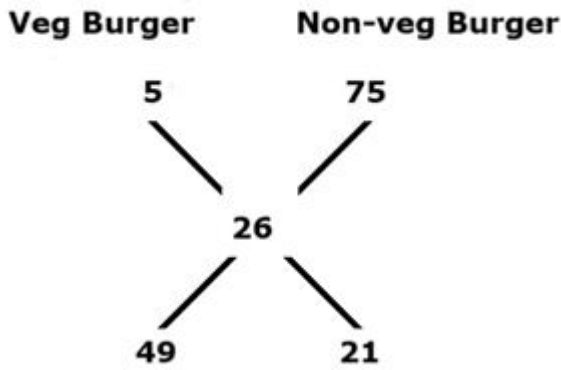
The total number of veg non-pizzas sold = $3/5$ of 50 = 30

So, the total revenue generated from all non-veg pizzas sold = $(60 * 30) = \text{Rs. } 1800$

Now,

The overall price of each burger (veg + non-veg) = $(1300 / 50) = 26$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg burgers sold = $49:21 = 7:3$

The total number of veg burgers sold = $7/10$ of $50 = 35$

So, the total revenue generated from the sale of all veg burgers = $(5 * 35) = \text{Rs. } 175$

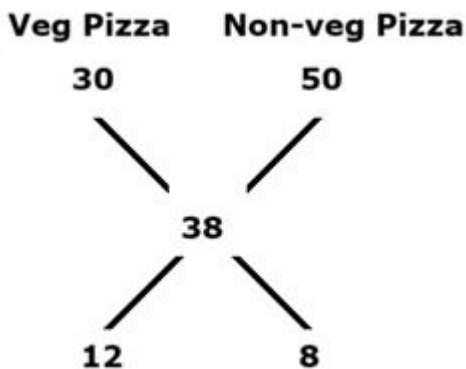
The total number of non-veg burgers sold = $3/10$ of $50 = 15$

So, the total revenue generated from the sale of all non-veg burgers = $(15 * 75) = \text{Rs. } 1125$

Shop C

The overall price of each pizza (veg + non-veg) = $(950 / 25) = \text{Rs. } 38$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg pizzas sold = $12:8 = 3:2$

The total number of veg pizzas sold = $3/5$ of $25 = 15$

So, the total revenue generated from all veg pizzas sold = $(30 * 15) = \text{Rs. } 450$

The total number of veg non-pizzas sold = $2/5$ of $25 = 10$

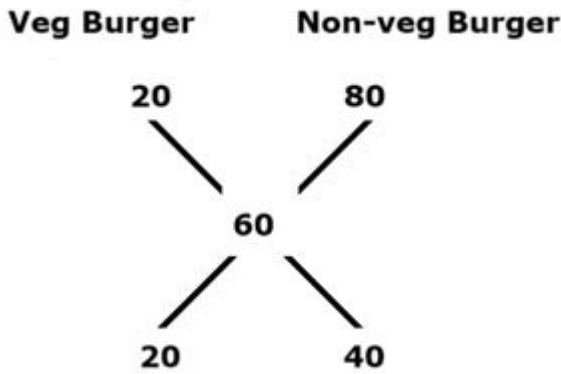
So, the total revenue generated from all non-veg pizzas sold = $(50 * 10) = \text{Rs. } 500$

Now,

The overall price of each burger (veg + non-veg) = $(3600 / 60) = 60$

So, using the allegation method, we get,

@QuizByMickey



So, the ratio of the number of veg and non-veg burgers sold = $20:40 = 1:2$

The total number of veg burgers sold = $1/3$ of $60 = 20$

So, the total revenue generated from the sale of all veg burgers = $(20 * 20) = \text{Rs. } 400$

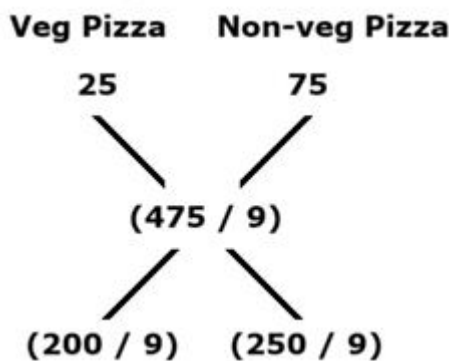
The total number of non-veg burgers sold = $2/3$ of $60 = 40$

So, the total revenue generated from the sale of all non-veg burgers = $(80 * 40) = \text{Rs. } 3200$

Shop D

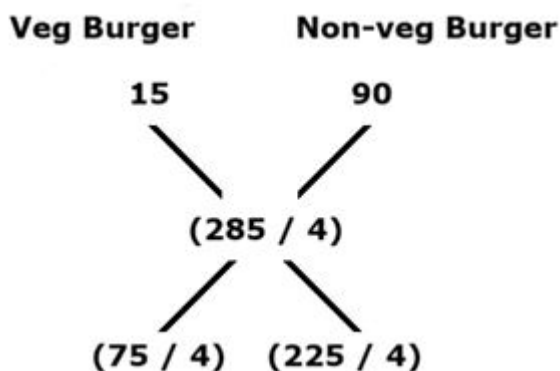
The overall price of each pizza (veg + non-veg) = $(4750 / 90) = \text{Rs. } (475 / 9)$

So, using the allegation method, we get,



So, the ratio of the number of veg and non-veg pizzas sold = $(200 / 9) : (250 / 9) = 200:250 = 4:5$

The total number of veg pizzas sold = $4/9$ of $90 = 40$



@QuizByMickey

So, the ratio of the number of veg and non-veg burgers sold = $(75 / 4) : (225 / 4) = 1:3$

The total number of veg burgers sold = $1/4$ of $40 = 10$

So, the total revenue generated from the sale of all veg burgers = $(15 * 10) = \text{Rs. } 150$

The total number of non-veg burgers sold = $3/4$ of $40 = 30$

So, the total revenue generated from the sale of all non-veg burgers = $(90 * 30) = \text{Rs. } 2700$

The total number of items sold

Shop	Veg Pizza	Non-veg Pizza	Veg Burger	Non-veg Burger
A	45	15	25	10
B	20	30	35	15
C	15	10	20	40
D	40	50	10	30
Total	130	105	90	95

The total revenue generated (in Rs.)

Shop	Veg Pizza	Non-veg Pizza	Veg Burger	Non-veg Burger
A	900	600	250	600
B	300	1800	175	1125
C	450	500	400	3200
D	1000	3750	150	2700
Total	2650	6650	975	7625

Answer: C

Shop E

The number of veg pizzas sold = 80% of $40 = 32$

The number of non-veg pizzas sold = $1/5$ of $50 = 10$

So, the total revenue generated from all veg pizzas sold = $(10 * 32) = \text{Rs. } 320$

And, the total revenue generated from all non-veg pizzas sold = $(20 * 10) = \text{Rs. } 200$

Therefore, the required sum = $(320 + 200) = \text{Rs. } 520$.

Hence, the correct answer is option C.

@QuizByMickey

7. Questions

Answer: C

Equation I: $A^2 - XA + 200 = 0$

The product of the roots = 200

The smaller root = 10 (second-lowest multiple of 5)

So, the greater root = $(200 / 10) = 20$

The sum of the roots = $(10 + 20) = 30$

$$X = 30$$

Equation II: $B^2 - YB - 84 = 0$

The product of the roots = -84

The positive root = 14 (exactly divisible by 7 but > 10 and < 20)

So, the negative root = $(-84 / 14) = -6$

The sum of the roots = $14 + (-6) = (14 - 6) = 8$

$$Y = 8$$

Equation III: $C^2 + ZC - 21 = 0$

The product of the roots = -21

The factors of 21 = 3 and 7

So, the greater root = -7

And, the smaller root = 3

The sum of the roots = $(-7 + 3) = -4$

$$Z = 4$$

The value of $(Y * Z - X) = (8 * 4 - 30) = (32 - 30) = 2$

2 is an even prime number.

Therefore, ? = even prime.

Hence, the correct answer is option C.

8. Questions

Answer: B

Case I

Let the possible values of P, R, and Q be 10, 11, and 12 (in any order)

$$Q = 12$$

$$R = 11$$

$$P = 10$$

Here, only one number is prime.

So, case I does not follow.

Case II

Let the possible values of P, R, and Q be 11, 12, and 13 (in any order)

$$Q = 13$$

@QuizByMickey

$$R = 12$$

$$P = 11$$

Here, 11 and 13 are prime numbers.

So, case II does follow.

Case III

Let the possible values of P, R, and Q be 12, 13, and 14 (in any order)

$$Q = 14$$

$$R = 13$$

$$P = 12$$

Here, only one number is prime.

So, case III does not follow.

Case IV

Let the possible values of P, R, and Q be 13, 14, and 15 (in any order)

$$Q = 15$$

$$R = 14$$

$$P = 13$$

Here, only one number is prime.

So, case IV does not follow.

Therefore, case II is satisfied.

So, the confirmed values of,

$$P = 11$$

$$Q = 13$$

$$R = 12$$

Equation I: $PX^2 - 41X + 2R = 0$

$$11X^2 - 41X + (2 * 12) = 0$$

$$11X^2 - 41X + 24 = 0$$

$$11X^2 - 33X - 8X + 24 = 0$$

$$11X(X - 3) - 8(X - 3) = 0$$

$$(11X - 8)(X - 3) = 0$$

$$X = +0.72 \text{ or } +3$$

@QuizByMickey

Equation II: $QY^2 - 43Y + R = 0$

$$13Y^2 - 43Y + 12 = 0$$

$$13Y^2 - 39Y - 4Y + 12 = 0$$

$$13Y(Y - 3) - 4(Y - 3) = 0$$

$$(13Y - 4)(Y - 3) = 0$$

$$Y = +0.30 \text{ or } +3$$

I: The greater root of equation I = 3

The greater root of equation II = 3

So, I is correct.

II: The smaller root of equation I = 0.72

0.72 is not more than 1.

So, II is not correct.

III: The difference between the roots of equation II = $(3 - 0.30) = 2.70$ (approx.)

2.70 is less than 3.

So, III is correct.

Therefore, both I and III are correct.

Hence, the correct answer is option B.

9. Questions

Answer: B

Equation I: $P^2 - 19P + (23C + 1) = 0$

The sum of the roots = 19

The smaller root = $(C + 2)$

The greater root = $19 - (C + 2) = (17 - C)$

The product of the roots = $(23C + 1)$

Now,

$$(17 - C) * (C + 2) = (23C + 1)$$

$$17C + 34 - C^2 - 2C = 23C + 1$$

$$15C + 34 - C^2 = 23C + 1$$

$$C^2 + 8C - 33 = 0$$

@QuizByMickey

$$C^2 + 11C - 3C - 33 = 0$$

$$C(C + 11) - 3(C + 11) = 0$$

$$(C - 3)(C + 11) = 0$$

$$C = +3 \text{ or } -11$$

So, $C = 3$ (positive integer)

Equation II: $Q^2 - 28Q + 33D = 0$

The sum of the roots = 28

The greater root = $(5D + 2)$

The smaller root = $28 - (5D + 2) = (26 - 5D)$

The product of the roots = $33D$

Now,

$$(5D + 2) * (26 - 5D) = 33D$$

$$130D - 25D^2 + 52 - 10D = 33D$$

$$120D - 25D^2 + 52 = 33D$$

$$25D^2 - 87D - 52 = 0$$

$$25D^2 - 100D + 13D - 52 = 0$$

$$25D(D - 4) + 13(D - 4) = 0$$

$$(25D + 13)(D - 4) = 0$$

$$D = -0.52 \text{ or } +4$$

So, $D = 4$ (positive integer)

Therefore, $C < D$.

Hence, the correct answer is option B.

10. Questions

Answer: D

The possible single-digit even composite number = 4, 6 or 8.

Case I

$$Y = 4^2 = 16$$

$$X + Y = Z$$

$$X + 16 = Z$$

@QuizByMickey

$$Z - X = 16 \text{ (not possible)}$$

Case II

$$Y = 6^2 = 36$$

$$X + Y = Z$$

$$X + 36 = Z$$

$$Z - X = 36 \text{ (possible)}$$

Case III

$$Y = 8^2 = 64$$

$$X + Y = Z$$

$$X + 64 = Z$$

$$Z - X = 64 \text{ (not possible)}$$

So, the confirmed value of $Y = 36$

$$Z - X = 36 \dots\dots\dots(i)$$

$$Z + X = (23 * 2)$$

$$Z + X = 46 \dots\dots\dots(ii)$$

By solving equations (i) and (ii), we get,

$$2X = 10$$

$$X = 5$$

Now, put the value of X in equation (i), we get,

$$Z - X = 36$$

$$Z - 5 = 36$$

$$Z = 41$$

Equation: $XT^2 + YT - Z = 0$

$$5T^2 + 36T - 41 = 0$$

$$5T^2 + 41T - 5T - 41 = 0$$

$$T(5T + 41) - 1(5T + 41) = 0$$

$$(T - 1)(5T + 41) = 0$$

$$T = +1 \text{ or } -41/5$$

$$T = +1 \text{ or } -8.20$$

So, the smaller root = -8.20 .

@QuizByMickey

Hence, the correct answer is option D.

11. Questions

Answer: E

Equation: $K^2 - 35K + (MN + 45) = 0$

The sum of the roots = 35

Case I

The smaller root = 2

The greater root = $(35 - 2) = 33$ (possible)

Case II

The smaller root = 3

The greater root = $(35 - 3) = 32$ (not possible)

Case III

The smaller root = 5

The greater root = $(35 - 5) = 30$ (not possible)

Case IV

The smaller root = 7

The greater root = $(35 - 7) = 28$ (not possible)

So, the confirmed roots are,

The smaller root = 2

The greater root = 33

So, the product of the roots = $(2 * 33) = 66$

$$MN + 45 = 66$$

$$MN = 21$$

The factors of 21 = 3 and 7

So, $M = 7$

And, $N = 3$

Therefore, the value of $(10M\% \text{ of } 20N) = (10 * 7)\% \text{ of } (20 * 3) = 70\% \text{ of } 60 = 42$.

Hence, the correct answer is option E.

12. Questions

2004 and 2019

The total number of voters registered in 2004 = $(16X + 80)$

@QuizByMickey

The number of votes cast in 2004 = 60% of $(16X + 80) = (9.60X + 48)$

The number of invalid votes cast in 2004 = $(9.60X + 48 - 900) = (9.60X - 852)$

The total number of voters registered in 2019 = 4000

The number of votes cast in 2019 = $(X - 70)\%$ of 4000 = $(40X - 2800)$

The number of invalid votes cast in 2019 = $(40X - 2800 - 1400) = (40X - 4200)$

Now,

$$(9.60X - 852) / (40X - 4200) = 1 / 2$$

$$19.20X - 1704 = 40X - 4200$$

$$20.80X = 2496$$

$$X = 120$$

2009 and 2014

The total number of voters registered in 2009 = 3200

The number of votes cast in 2009 = $(4Y - 5)\%$ of 3200 = $(128Y - 160)$

The number of non-cast votes in 2009 = $3200 - (128Y - 160) = (3360 - 128Y)$

The total number of voters registered in 2014 = 80Y

The number of votes cast in 2014 = 80% of 80Y = 64Y

The number of non-cast votes in 2014 = $(80Y - 64Y) = 16Y$

Now,

$$(3360 - 128Y) - 16Y = 480$$

$$3360 - 144Y = 480$$

$$144Y = 2880$$

$$Y = 20$$

Now, the number of invalid votes cast in 2009 = $(128Y - 160) - (Z + 300) = (128 * 20 - 160) - (Z + 300)$
 $= (2560 - 160) - (Z + 300) = 2400 - (Z + 300) = (2100 - Z)$

And, the number of invalid votes cast in 2014 = $(64Y - W) = (64 * 20 - W) = (1280 - W)$

So,

$$(2100 - Z) + (1280 - W) = (440 * 2)$$

$$3380 - Z - W = 880$$

$$Z + W = 2500 \dots \dots \dots (I)$$

Also,

$$Z - W = (7X + 3Y)$$

$$Z - W = (7 * 120 + 3 * 20)$$

$$Z - W = (840 + 60)$$

$$Z - W = 900 \dots \dots \dots (II)$$

By solving equations (I) and (II), we get,

$$2W = 1600$$

$$W = 800$$

Now, put the value of W in equation (I), we get,

$$Z + W = 2500$$

$$Z + 800 = 2500$$

$$Z = 1700$$

2004

The total number of registered votes = $(16X + 80) = (16 * 120 + 80) = (1920 + 80) = 2000$

The number of votes cast = 60% of 2000 = 1200

The number of non-cast votes = $(2000 - 1200) = 800$

The number of valid votes cast = 900

The number of invalid votes cast = $(1200 - 900) = 300$

2009

The total number of registered votes = 3200

The number of votes cast = $(4Y - 5)\%$ of 3200 = $(4 * 20 - 5)\%$ of 3200 = $(80 - 5)\%$ of 3200 = 75% of 3200 = 2400

The number of non-cast votes = $(3200 - 2400) = 800$

The number of valid votes cast = $(Z + 300) = (1700 + 300) = 2000$

The number of invalid votes cast = $(2400 - 2000) = 400$

2014

The total number of registered votes = $80Y = (80 * 20) = 1600$ @QuizByMickey

The number of votes cast = 80% of 1600 = 1280

The number of non-cast votes = $(1600 - 1280) = 320$

The number of valid votes cast = $W = 800$

The number of invalid votes cast = $(1280 - 800) = 480$

2019

The total number of registered votes = 4000

The number of votes cast = $(X - 70)\%$ of 4000 = $(120 - 70)\%$ of 4000 = 50% of 4000 = 2000

The number of non-cast votes = $(4000 - 2000) = 2000$

The number of valid votes cast = 1400

The number of invalid votes cast = $(2000 - 1400) = 600$

Year	The total number of voters registered	The number of votes cast	The number of non-cast votes	The number of valid votes cast	The number of invalid votes cast
2004	2000	1200	800	900	300
2009	3200	2400	800	2000	400
2014	1600	1280	320	800	480
2019	4000	2000	2000	1400	600

Answer: E

Now, according to the question,

2009

The total number of votes cast = 2400

The total number of votes cast by females = $1/8$ of 2400 = 300

The total number of votes cast by males = $(2400 - 300) = 2100$

The total number of valid votes cast = 2000

The number of valid votes cast by males = $9/10$ of 2000 = 1800

The number of valid votes cast by females = $1/10$ of 2000 = 200

The number of invalid votes cast by males = $(2100 - 1800) = 300$

The number of invalid votes cast by females = $(300 - 200) = 100$

Therefore, the required difference = $(300 - 100) = 200$.

Hence, the correct answer is option E.

13. Questions

2004 and 2019

The total number of voters registered in 2004 = $(16X + 80)$

The number of votes cast in 2004 = 60% of $(16X + 80) = (9.60X + 48)$

The number of invalid votes cast in 2004 = $(9.60X + 48 - 900) = (9.60X - 852)$

The total number of voters registered in 2019 = 4000

The number of votes cast in 2019 = $(X - 70)\%$ of 4000 = $(40X - 2800)$

@QuizByMickey

The number of invalid votes cast in 2019 = $(40X - 2800 - 1400) = (40X - 4200)$

Now,

$$(9.60X - 852) / (40X - 4200) = 1 / 2$$

$$19.20X - 1704 = 40X - 4200$$

$$20.80X = 2496$$

$$X = 120$$

2009 and 2014

The total number of voters registered in 2009 = 3200

The number of votes cast in 2009 = $(4Y - 5)\%$ of 3200 = $(128Y - 160)$

The number of non-cast votes in 2009 = $3200 - (128Y - 160) = (3360 - 128Y)$

The total number of voters registered in 2014 = $80Y$

The number of votes cast in 2014 = 80% of $80Y = 64Y$

The number of non-cast votes in 2014 = $(80Y - 64Y) = 16Y$

Now,

$$(3360 - 128Y) - 16Y = 480$$

$$3360 - 144Y = 480$$

$$144Y = 2880$$

$$Y = 20$$

Now, the number of invalid votes cast in 2009 = $(128Y - 160) - (Z + 300) = (128 * 20 - 160) - (Z + 300)$
 $= (2560 - 160) - (Z + 300) = 2400 - (Z + 300) = (2100 - Z)$

And, the number of invalid votes cast in 2014 = $(64Y - W) = (64 * 20 - W) = (1280 - W)$

So,

$$(2100 - Z) + (1280 - W) = (440 * 2)$$

$$3380 - Z - W = 880$$

$$Z + W = 2500 \dots \dots \dots (I)$$

Also,

$$Z - W = (7X + 3Y)$$

$$Z - W = (7 * 120 + 3 * 20)$$

$$Z - W = (840 + 60)$$

$$Z - W = 900 \dots \dots \dots (II)$$

By solving equations (I) and (II), we get,

@QuizByMickey

$$2W = 1600$$

$$W = 800$$

Now, put the value of W in equation (I), we get,

$$Z + W = 2500$$

$$Z + 800 = 2500$$

$$Z = 1700$$

2004

The total number of registered votes = $(16X + 80) = (16 * 120 + 80) = (1920 + 80) = 2000$

The number of votes cast = 60% of 2000 = 1200

The number of non-cast votes = $(2000 - 1200) = 800$

The number of valid votes cast = 900

The number of invalid votes cast = $(1200 - 900) = 300$

2009

The total number of registered votes = 3200

The number of votes cast = $(4Y - 5)\%$ of 3200 = $(4 * 20 - 5)\%$ of 3200 = $(80 - 5)\%$ of 3200 = 75% of 3200 = 2400

The number of non-cast votes = $(3200 - 2400) = 800$

The number of valid votes cast = $(Z + 300) = (1700 + 300) = 2000$

The number of invalid votes cast = $(2400 - 2000) = 400$

2014

The total number of registered votes = $80Y = (80 * 20) = 1600$

The number of votes cast = 80% of 1600 = 1280

The number of non-cast votes = $(1600 - 1280) = 320$

The number of valid votes cast = $W = 800$

The number of invalid votes cast = $(1280 - 800) = 480$

@QuizByMickey

2019

The total number of registered votes = 4000

The number of votes cast = $(X - 70)\%$ of 4000 = $(120 - 70)\%$ of 4000 = 50% of 4000 = 2000

The number of non-cast votes = $(4000 - 2000) = 2000$

The number of valid votes cast = 1400

The number of invalid votes cast = $(2000 - 1400) = 600$

Year	The total number of voters registered	The number of votes cast	The number of non-cast votes	The number of valid votes cast	The number of invalid votes cast
2004	2000	1200	800	900	300
2009	3200	2400	800	2000	400
2014	1600	1280	320	800	480
2019	4000	2000	2000	1400	600

Answer: D

The number of votes cast in 2014 = 1280

The number of votes cast in 2019 = 2000

Therefore, the required average = $(1280 + 2000) / 2 = (3280 / 2) = 1640$.

Hence, the correct answer is option D.

14. Questions

2004 and 2019

The total number of voters registered in 2004 = $(16X + 80)$

The number of votes cast in 2004 = 60% of $(16X + 80) = (9.60X + 48)$

The number of invalid votes cast in 2004 = $(9.60X + 48 - 900) = (9.60X - 852)$

The total number of voters registered in 2019 = 4000

The number of votes cast in 2019 = $(X - 70)\%$ of 4000 = $(40X - 2800)$

The number of invalid votes cast in 2019 = $(40X - 2800 - 1400) = (40X - 4200)$

Now,

$$(9.60X - 852) / (40X - 4200) = 1 / 2$$

$$19.20X - 1704 = 40X - 4200$$

$$20.80X = 2496$$

$$X = 120$$

@QuizByMickey

2009 and 2014

The total number of voters registered in 2009 = 3200

The number of votes cast in 2009 = $(4Y - 5)\%$ of 3200 = $(128Y - 160)$

The number of non-cast votes in 2009 = $3200 - (128Y - 160) = (3360 - 128Y)$

The total number of voters registered in 2014 = 80Y

The number of votes cast in 2014 = 80% of 80Y = 64Y

The number of non-cast votes in 2014 = $(80Y - 64Y) = 16Y$

Now,

$$(3360 - 128Y) - 16Y = 480$$

$$3360 - 144Y = 480$$

$$144Y = 2880$$

$$Y = 20$$

$$\begin{aligned} \text{Now, the number of invalid votes cast in 2009} &= (128Y - 160) - (Z + 300) = (128 * 20 - 160) - (Z + 300) \\ &= (2560 - 160) - (Z + 300) = 2400 - (Z + 300) = (2100 - Z) \end{aligned}$$

$$\text{And, the number of invalid votes cast in 2014} = (64Y - W) = (64 * 20 - W) = (1280 - W)$$

So,

$$(2100 - Z) + (1280 - W) = (440 * 2)$$

$$3380 - Z - W = 880$$

$$Z + W = 2500 \dots \dots \dots \text{(I)}$$

Also,

$$Z - W = (7X + 3Y)$$

$$Z - W = (7 * 120 + 3 * 20)$$

$$Z - W = (840 + 60)$$

$$Z - W = 900 \dots \dots \dots \text{(II)}$$

By solving equations (I) and (II), we get,

$$2W = 1600$$

$$W = 800$$

Now, put the value of W in equation (I), we get,

$$Z + W = 2500$$

$$Z + 800 = 2500$$

$$Z = 1700$$

2004

$$\text{The total number of registered votes} = (16X + 80) = (16 * 120 + 80) = (1920 + 80) = 2000$$

$$\text{The number of votes cast} = 60\% \text{ of } 2000 = 1200$$

$$\text{The number of non-cast votes} = (2000 - 1200) = 800$$

$$\text{The number of valid votes cast} = 900$$

$$\text{The number of invalid votes cast} = (1200 - 900) = 300$$

@QuizByMickey

2009

The total number of registered votes = 3200

The number of votes cast = $(4Y - 5)\%$ of 3200 = $(4 * 20 - 5)\%$ of 3200 = $(80 - 5)\%$ of 3200 = 75% of 3200 = 2400

The number of non-cast votes = $(3200 - 2400) = 800$

The number of valid votes cast = $(Z + 300) = (1700 + 300) = 2000$

The number of invalid votes cast = $(2400 - 2000) = 400$

2014

The total number of registered votes = $80Y = (80 * 20) = 1600$

The number of votes cast = 80% of 1600 = 1280

The number of non-cast votes = $(21600 - 1280) = 320$

The number of valid votes cast = $W = 800$

The number of invalid votes cast = $(1280 - 800) = 480$

2019

The total number of registered votes = 4000

The number of votes cast = $(X - 70)\%$ of 4000 = $(120 - 70)\%$ of 4000 = 50% of 4000 = 2000

The number of non-cast votes = $(4000 - 2000) = 2000$

The number of valid votes cast = 1400

The number of invalid votes cast = $(2000 - 1400) = 600$

Year	The total number of voters registered	The number of votes cast	The number of non-cast votes	The number of valid votes cast	The number of invalid votes cast
2004	2000	1200	800	900	300
2009	3200	2400	800	2000	400
2014	1600	1280	320	800	480
2019	4000	2000	2000	1400	600

Answer: A

2024

The number of votes cast = 840

The number of invalid votes cast = $5/14$ of 840 = 300

So, the number of non-cast votes = $(300 + 60) = 360$

Now, the total number of votes registered in 2024 = $(840 + 360) = 1200$

The total number of votes registered in 2004 = 2000

Therefore, $? = (2000 - 1200) / 2000 * 100 = 800/2000 * 100 = 40\%$.

Hence, the correct answer is option A.

15. Questions

2004 and 2019

The total number of voters registered in 2004 = $(16X + 80)$

The number of votes cast in 2004 = 60% of $(16X + 80) = (9.60X + 48)$

The number of invalid votes cast in 2004 = $(9.60X + 48 - 900) = (9.60X - 852)$

The total number of voters registered in 2019 = 4000

The number of votes cast in 2019 = $(X - 70)\%$ of 4000 = $(40X - 2800)$

The number of invalid votes cast in 2019 = $(40X - 2800 - 1400) = (40X - 4200)$

Now,

$$(9.60X - 852) / (40X - 4200) = 1 / 2$$

$$19.20X - 1704 = 40X - 4200$$

$$20.80X = 2496$$

$$X = 120$$

2009 and 2014

The total number of voters registered in 2009 = 3200

The number of votes cast in 2009 = $(4Y - 5)\%$ of 3200 = $(128Y - 160)$

The number of non-cast votes in 2009 = $3200 - (128Y - 160) = (3360 - 128Y)$

The total number of voters registered in 2014 = 80Y

The number of votes cast in 2014 = 80% of 80Y = 64Y

The number of non-cast votes in 2014 = $(80Y - 64Y) = 16Y$

Now,

$$(3360 - 128Y) - 16Y = 480$$

$$3360 - 144Y = 480$$

$$144Y = 2880$$

$$Y = 20$$

Now, the number of invalid votes cast in 2009 = $(128Y - 160) - (Z + 300) = (128 * 20 - 160) - (Z + 300)$
 $= (2560 - 160) - (Z + 300) = 2400 - (Z + 300) = (2100 - Z)$

And, the number of invalid votes cast in 2014 = $(64Y - W) = (64 * 20 - W) = (1280 - W)$

@QuizByMickey

So,

$$(2100 - Z) + (1280 - W) = (440 * 2)$$

$$3380 - Z - W = 880$$

$$Z + W = 2500 \dots \dots \dots (I)$$

Also,

$$Z - W = (7X + 3Y)$$

$$Z - W = (7 * 120 + 3 * 20)$$

$$Z - W = (840 + 60)$$

$$Z - W = 900 \dots \dots \dots (II)$$

By solving equations (I) and (II), we get,

$$2W = 1600$$

$$W = 800$$

Now, put the value of W in equation (I), we get,

$$Z + W = 2500$$

$$Z + 800 = 2500$$

$$Z = 1700$$

2004

$$\text{The total number of registered votes} = (16X + 80) = (16 * 120 + 80) = (1920 + 80) = 2000$$

$$\text{The number of votes cast} = 60\% \text{ of } 2000 = 1200$$

$$\text{The number of non-cast votes} = (2000 - 1200) = 800$$

$$\text{The number of valid votes cast} = 900$$

$$\text{The number of invalid votes cast} = (1200 - 900) = 300$$

2009

$$\text{The total number of registered votes} = 3200$$

$$\text{The number of votes cast} = (4Y - 5)\% \text{ of } 3200 = (4 * 20 - 5)\% \text{ of } 3200 = (80 - 5)\% \text{ of } 3200 = 75\% \text{ of } 3200 = 2400$$

$$\text{The number of non-cast votes} = (3200 - 2400) = 800$$

$$\text{The number of valid votes cast} = (Z + 300) = (1700 + 300) = 2000$$

$$\text{The number of invalid votes cast} = (2400 - 2000) = 400$$

2014

$$\text{The total number of registered votes} = 80Y = (80 * 20) = 1600$$

The number of votes cast = 80% of 1600 = 1280

The number of non-cast votes = (21600 – 1280) = 320

The number of valid votes cast = W = 800

The number of invalid votes cast = (1280 – 800) = 480

2019

The total number of registered voters = 4000

The number of votes cast = (X – 70)% of 4000 = (120 – 70)% of 4000 = 50% of 4000 = 2000

The number of non-cast votes = (4000 – 2000) = 2000

The number of valid votes cast = 1400

The number of invalid votes cast = (2000 – 1400) = 600

Year	The total number of voters registered	The number of votes cast	The number of non-cast votes	The number of valid votes cast	The number of invalid votes cast
2004	2000	1200	800	900	300
2009	3200	2400	800	2000	400
2014	1600	1280	320	800	480
2019	4000	2000	2000	1400	600

Answer: B

I: $(W / Y) = (800 / 20) = 40$

So, I is correct.

II: $(X + 1580) = (120 + 1580) = 1700$

Z = 1700

So, II is correct.

III: $(W - X) = (800 - 120) = 680$

$(Z / Y) = (1700 / 20) = 85$

$680 \neq 85$

So, III is not correct.

Therefore, both I and II are correct.

Hence, the correct answer is option B.

16. Questions

Answer: C

@QuizByMickey

Series I: 24, 35, 13, 46, __[P]__, 57

$$1^{\text{st}} \text{ term} = 24$$

$$2^{\text{nd}} \text{ term} = (24 + 11) = 35$$

$$3^{\text{rd}} \text{ term} = (35 - 22) = 13$$

$$4^{\text{th}} \text{ term} = (13 + 33) = 46$$

$$5^{\text{th}} \text{ term} = (46 - 44) = 2 \text{ (P)}$$

$$6^{\text{th}} \text{ term} = (2 + 55) = 57$$

So, the missing term = P = 2

Series II: 14, 29, 89, __[Q]__, 1799, 10799

$$1^{\text{st}} \text{ term} = 14$$

$$2^{\text{nd}} \text{ term} = (14 * 2 + 1) = (28 + 1) = 29$$

$$3^{\text{rd}} \text{ term} = (29 * 3 + 2) = (87 + 2) = 89$$

$$4^{\text{th}} \text{ term} = (89 * 4 + 3) = (356 + 3) = 359 \text{ (Q)}$$

$$5^{\text{th}} \text{ term} = (359 * 5 + 4) = (1795 + 4) = 1799$$

$$6^{\text{th}} \text{ term} = (1799 * 6 + 5) = (10794 + 5) = 10799$$

So, the missing term = Q = 359

$$\text{I: } (P + Q) = (2 + 359) = 361$$

$$\sqrt{361} = 19$$

So, I is correct.

II: Only Q (359) is a three-digit natural number.

So, II is not correct.

III: P (2) is an even prime number.

So, III is correct.

Therefore, both I and III are correct.

Hence, the correct answer is option C.

17. Questions

Answer: B

Series I: 7, 11, 36, 81, 221, 417

@QuizByMickey

$$1^{\text{st}} \text{ term} = 7$$

$$2^{\text{nd}} \text{ term} = (7 + 2^2) = (7 + 4) = 11$$

$$3^{\text{rd}} \text{ term} = (11 + 5^2) = (11 + 25) = 36$$

$$4^{\text{th}} \text{ term} = (36 + 8^2) = (36 + 64) = 100 \text{ (81 is wrong)}$$

$$5^{\text{th}} \text{ term} = (100 + 11^2) = (100 + 121) = 221$$

$$6^{\text{th}} \text{ term} = (221 + 14^2) = (221 + 196) = 417$$

So, the wrong term = 81

And, the correct term = 100

Series II:

$$1^{\text{st}} \text{ term} = A = 100$$

$$2^{\text{nd}} \text{ term} = (100 + 2^2) = (100 + 4) = 104$$

$$3^{\text{rd}} \text{ term} = (104 + 5^2) = (104 + 25) = 129$$

$$4^{\text{th}} \text{ term} = (129 + 8^2) = (129 + 64) = 193$$

$$5^{\text{th}} \text{ term} = (193 + 11^2) = (193 + 121) = 314$$

$$6^{\text{th}} \text{ term} = (314 + 14^2) = (314 + 196) = 510$$

So, B = 510

Equation: $L^2 - (\sqrt{A} + 1)L + (B / 17) = 0$

$$L^2 - (\sqrt{100} + 1)L + (510 / 17) = 0$$

$$L^2 - (10 + 1)L + 30 = 0$$

$$L^2 - 11L + 30 = 0$$

$$L^2 - 6L - 5L + 30 = 0$$

$$L(L - 6) - 5(L - 6) = 0$$

$$(L - 5)(L - 6) = 0$$

$$L = +5 \text{ or } +6$$

Therefore, the greater root of the equation = 6.

Hence, the correct answer is option B.

18. Questions

@QuizByMickey

Answer: C

Series I: 24, 144, 16, 72, 6, 72

1st term = 24

2nd term = $(24 * 6) = 144$

3rd term = $(144 / 12) = 12$ (16 is first wrong term)

4th term = $(12 * 6) = 72$

5th term = $(72 / 12) = 6$

6th term = $(6 * 6) = 36$ (72 is second wrong term)

So, the first wrong term = 16

And, the second wrong term = 72

The first correct term = 12

The second correct term = 36

Therefore, $X = 12$

And, $Y = 36$

Series II: $[X + Y]$, 24, 36, 90, 315, __[?]__

$[12 + 36]$, 24, 36, 90, 315, __[?]__

48, 24, 36, 90, 315, __[?]__

1st term = 48

2nd term = $(48 * 0.50) = 24$

3rd term = $(24 * 1.50) = 36$

4th term = $(36 * 2.50) = 90$

5th term = $(90 * 3.50) = 315$

6th term = $(315 * 4.50) = 1417.50$

Therefore, $? = 1417.50$.

Hence, the correct answer is option C.

19. Questions

Answer: E

C = 9 (highest single-digit perfect square)

@QuizByMickey

$$B + A = 5C$$

$$B + A = (5 * 9)$$

$$B + A = 45 \dots\dots\dots(i)$$

Also,

$$B - A = 13 \dots\dots\dots(ii)$$

By solving equations (i) and (ii), we get,

$$2A = 32$$

$$A = 16$$

Now, put the value of A in equation (i), we get,

$$B + A = 45$$

$$B + 16 = 45$$

$$B = 29$$

Series I: [A], [3C + 2], 55, 94, 145, 211

$$16, [3 * 9 + 2], 55, 94, 145, 211$$

$$16, [27 + 2], 55, 94, 145, 211$$

$$16, 29, 55, 94, 145, 211$$

$$1^{\text{st}} \text{ term} = 16$$

$$2^{\text{nd}} \text{ term} = (16 + 13) = 29$$

$$3^{\text{rd}} \text{ term} = (29 + 26) = 55$$

$$4^{\text{th}} \text{ term} = (55 + 39) = 94$$

$$5^{\text{th}} \text{ term} = (94 + 52) = 146 \text{ (145 is wrong)}$$

$$6^{\text{th}} \text{ term} = (146 + 65) = 211$$

So, the wrong term = 145

And, the correct term = 146

Series II: [B - 20], 22, 48, 87, 139, 200

$$[29 - 20], 22, 48, 87, 139, 200$$

$$9, 22, 48, 87, 139, 200$$

$$1^{\text{st}} \text{ term} = 9$$

$$2^{\text{nd}} \text{ term} = (9 + 13) = 22$$

@QuizByMickey

$$3^{\text{rd}} \text{ term} = (22 + 26) = 48$$

$$4^{\text{th}} \text{ term} = (48 + 39) = 87$$

$$5^{\text{th}} \text{ term} = (87 + 52) = 139$$

$$6^{\text{th}} \text{ term} = (139 + 65) = 204 \text{ (200 is wrong)}$$

So, the wrong term = 200

And, the correct term = 204

Therefore, the required difference = $(204 - 146) = 58$.

Hence, the correct answer is option E.

20. Questions

Answer: A

Series I: 100, 127, 2, __[M]__, - 384, 947

$$1^{\text{st}} \text{ term} = 100$$

$$2^{\text{nd}} \text{ term} = (100 + 3^3) = (100 + 27) = 127$$

$$3^{\text{rd}} \text{ term} = (127 - 5^3) = (127 - 125) = 2$$

$$4^{\text{th}} \text{ term} = (2 + 7^3) = (2 + 343) = 345 \text{ (M)}$$

$$5^{\text{th}} \text{ term} = (345 - 9^3) = (345 - 729) = - 384$$

$$6^{\text{th}} \text{ term} = (- 384 + 11^3) = (- 384 + 1331) = 947$$

So, the missing term = M = 345

Series II: [M / 3], 140, 17, 360, - 369, 962

[345 / 3], 140, 17, 360, - 369, 962

115, 140, 17, 360, - 369, 962

$$1^{\text{st}} \text{ term} = 115$$

$$2^{\text{nd}} \text{ term} = (115 + 3^3) = (115 + 27) = 142 \text{ (140 is wrong term)}$$

$$3^{\text{rd}} \text{ term} = (142 - 5^3) = (142 - 125) = 17$$

$$4^{\text{th}} \text{ term} = (17 + 7^3) = (17 + 343) = 360$$

$$5^{\text{th}} \text{ term} = (360 - 9^3) = (360 - 729) = - 369$$

$$6^{\text{th}} \text{ term} = (- 369 + 11^3) = (- 369 + 1331) = 962$$

@QuizByMickey

So, the wrong term = $N = 140$

The correct term = 142

Therefore, $M > N$.

Hence, the correct answer is option A.

@QuizByMickey